A Coarse-to-Fine Semantic Conflict Detection System for Ex-Ante Davao City Ordinances Using Information Retrieval and Natural Language Inference

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April 2026

Acknowledgement

Writing the first half of this thesis was a massive undertaking, and we could not have crossed the finish line without the people who backed us up along the way.

To our professor, Ma'am Grace Tacadao, thank you for believing in us from day one. You took a chance by letting us be the only pair in the class, and you understood our vision immediately with our topic. Thank you for being so supportive and for being the exact kind of constructive critic we needed to actually make this paper work.

To our adviser, Sir Adrian "Ogs" Ablazo, thank you for saying yes the second we pitched this idea to you. Your chill energy was exactly what we needed to keep our sanity intact. Thank you for being incredibly receptive, for showing us exactly what was missing, and for being a safe space where we could share both our exciting progress and our random frustrations.

To the staff at the Davao City Sangguniang Panlungsod, especially Mr. Kevin B. Alfoja and Ms. Ma. Theresa A. Reyes, thank you for patiently answering our emails, catering to our concerns, and ultimately trusting us with the LISSP dataset. This system literally would not exist without your help.

To our defense panelists, thank you for your sharp, kind critiques and for pushing us to bring this vision to life. (And an advance thank you for hopefully not bombarding us with too many hard questions during the actual defense!)

To our families, friends, and classmates—thank you for the constant support, the patience when we were busy making this paper happen, and for believing we could survive our academics and this thesis. A special shoutout to Vince Demeer Valmores for the extra patience and encouragement behind the scenes.

Above all, to the Almighty God/Allah SWT, the ultimate source of strength, wisdom, and grace. Thank you for making all of this possible and for seeing us through to a successful end.

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Chapter 1

Introduction

1.1 Background of the Study

A functional legal system requires strict consistency so that citizens can clearly understand their rights and obligations [36]. However, the sheer volume of modern legislation has grown so rapidly that human practitioners can no longer track every active rule [109]. As governing bodies continuously pass new regulations to address emerging issues, they frequently enact laws that inadvertently contradict older ones [64]. Because the total number of documents is overwhelming, identifying these hidden conflicts through manual review is computationally impossible for legal staff. This issue is particularly severe in local government units, where historical ordinances are often scattered across fragmented files rather than maintained in a structured digital database [32].

Within the Philippine legal framework, local governments like Davao City operate under delegated authority and can only enact laws permitted by the national government [43]. Due to this structural limitation, the Magtajas v. Pryce Doctrine strictly prohibits any local municipal ordinance from contradicting established national statutes [123]. Consequently, in the local context, city councils must rigorously audit every newly proposed draft ordinance to ensure it logically aligns with existing laws before final approval. If reviewers overlook a normative conflict, such as a local ordinance prohibiting an activity that a national law explicitly permits, the resulting local legislation is legally void. Passing invalid laws creates severe jurisdictional disputes, wastes administrative resources, and ultimately compromises the constitutional rights of the public.

To address these legal risks, the legislative process must utilize an ex-ante evaluation: a term derived from Latin meaning 'before the event.' Unlike an 'ex-post' review, which happens after a law is already in effect and often after legal damage has occurred, an ex-ante audit acts as a first line of defense. By checking a proposed ordinance for conflicts while it is still in the drafting phase, the Sangguniang Panlungsod can proactively align local rules with national statutes. This preventative approach is consistent with modern Philippine standards like Republic Act No. 11032 or the Ease of Doing Business and Efficient Government Service Delivery Act of 2018, which encourages government agencies to perform regulatory assessments to ensure that new rules do not create unnecessary or conflicting burdens on the public.

The decentralized drafting process of local ordinances further complicates manual review because the resulting texts are often highly unstructured [112]. City councils frequently splice and amend different drafts, producing verbose documents that blend formal legal English with regional dialects like Cebuano and administrative "Taglish" [33]. Furthermore, because municipal archives heavily rely on digitized PDF scans, the text suffers from severe Optical Character Recognition (OCR) errors. While OCR is typically 99 percent accurate for clean English text, studies show accuracy can drop to 78 percent for low-resource languages, with Cebuano specifically seeing word error rates around 6.9 percent [96]. These errors cause an “error cascade” that shatters standard vocabulary, making it difficult for traditional search engines to semantically differentiate between a statutory obligation and a strict prohibition.

In light of these challenges, the proponents seek to develop a specialized application capable of retrieving relevant legal documents and comparing them against proposed ordinances to detect potential conflicts.

1.2 Problem Statement

This study is conducted to investigate the use of a multi-stage architecture to automate the manual auditing of proposed local legislation in Davao City.

Specifically, the study aims to answer the following research questions:

- What are the specific linguistic parameters and structural markers that define a "semantic conflict" between a proposed local ordinance and existing local or national laws?
- How can a Coarse-to-Fine system be designed to efficiently retrieve relevant national and local statutes before performing deep logical inference on constrained LGU hardware?
- Which combination of a hybrid Information Retrieval (IR) algorithms and Natural Language Inference (NLI) models provides the highest accuracy when processing the specific hierarchy of Philippine legal texts?
- What is the empirical performance of the proposed pipeline when evaluated against a manually curated Ground Truth dataset of local and national conflict pairs?
- How usable and viable is the final system interface for key stakeholders within the Davao City Sangguniang Panlungsod?

1.3 Objectives of the Study

The primary goal of this research is to develop a computationally efficient, two-stage semantic conflict detection system for proposed Davao City ordinances.

Specifically, this study aims to:

- Identify the specific linguistic characteristics and structural constraints of Davao City's legislative archive that contribute to semantic ambiguity and retrieval failure;
- Design a Coarse-to-Fine computational pipeline that decouples document retrieval from logical inference to optimize resource usage for local deployment;
- Implement a hybrid Information Retrieval (IR) retrieval algorithm and Natural Language Inference (NLI) system specifically calibrated for the Philippine legal hierarchy;
- Evaluate the system's reliability by comparing its automated detections against a verified set of known legal conflicts to measure how accurately it identifies contradictions; and
- Develop a user-friendly web interface for the Davao City Sangguniang Panlungsod that primarily functions as accepting uploading drafts and providing real-time conflict alerts.

1.4 Significance of the Study

The Davao City Sangguniang Panlungsod stands as the primary beneficiary of this automated consistency checking system. By transitioning from manual review to computational auditing, the tool saves legislative staff hundreds of hours of reading time. This technological intervention actively prevents the city from enacting logically flawed laws that invite future litigation or administrative deadlocks. It empowers the local government to draft regulatory measures with a significantly higher degree of legal certainty and operational speed. Ultimately, the system modernizes the legislative lifecycle by embedding rigorous logical checks directly into the initial drafting phase.

This study also provides substantial value to legal researchers and the local Natural Language Processing community. It establishes an empirical baseline for processing messy, localized Philippine legal corpora that frequently contain "Taglish" and archaic Spanish loanwords. The research demonstrates that efficient, edge-deployable encoder models can successfully parse OCR-degraded documents without relying on expensive, cloud-based supercomputers. Beyond academic and governmental applications, the general public benefits directly through the consistent implementation of coherent public policy. The tool acts as a procedural safety net, protecting citizens from the legal traps and frustrations of contradictory municipal obligations.

1.5 Scope and Limitations

The scope of this study focuses on the development of a semantic conflict detection system designed to compare local ordinances from Davao City against a static database of over 10,000 national statutes, specifically Republic Acts, Acts, Batas Pambansa, and Commonwealth Acts. To keep the research manageable, other legal documents such as executive issuances (Presidential Decrees) and judicial decisions are excluded.

A significant limitation is the incompleteness of the municipal archive. For ordinances enacted earlier than 2021, the collection consists primarily of “Landmark Ordinances” rather than the full set of laws for those years. While the proponents utilized full access to post-2021 records via the LISSP database, gaps still exist in the numerical sequence of the documents, resulting in a final knowledge base of approximately 1,660 ordinances. These records represent the most complete dataset currently available after exhausting all retrieval efforts through public repositories and direct coordination with the Sangguniang Panlungsod; consequently, the system’s ability to detect conflicts is limited to the records present in this compiled database [113].

Furthermore, the system is restricted to English-language texts to avoid the complexities of translation and regional dialects. This choice is necessary because local municipal records are currently under-represented in legal AI research; while national laws are well-studied in central databases, local ordinances remain neglected because they are unorganized and often “messy” [113, 6, 84]. This has led to a lack of a computational footprint, meaning there are no high-quality, standardized datasets for local ordinances that an AI can follow to learn legal reasoning [15]. Additionally, focusing only on English avoids “semantic drift,” where the specific legal meaning of a rule may get lost or accidentally changed when moved into a local dialect [67]. The database is compiled from existing public government repositories, meaning the tool will not perform live web scraping during its operation.

Operationally, the system uses a Natural Language Inference (NLI) framework to perform a single-premise comparison, evaluating a draft ordinance against a single national provision at a time. While the system uses Asymmetric Span-Level Pairing to compare a multi-sentence block from a local draft against a specific national rule, it does not perform “multi-hop” reasoning. This means it cannot identify conflicts that only appear when combining two or more separate national laws, as such complex processing would exceed the hardware limits of a standard office computer. The system classifies these relationships as Entailment, Neutral, or Contradiction, with a specific optimization for flagging contradictions.

To ensure transparency, the study uses a “white box” approach to Explainable AI (XAI), providing mathematical data like confidence scores and attention weights instead of generated text summaries. To conserve resources and ensure practical deployment, the framework deliberately excludes generative Large Language Models (LLMs); it cannot write, rewrite, or automatically correct legal text. Finally, all development and testing are strictly limited to a single consumer-grade device. This ensures the final product is a practical diagnostic tool that can be deployed by a standard Local Government Unit (LGU) without the need for expensive, enterprise-grade cloud computing resources.

1.6 Definition of Terms

Coarse-to-Fine Architecture. A two-step computer process. First, the system quickly searches through thousands of documents to find ”candidates” (coarse), then it carefully analyzes those specific candidates for logical conflicts (fine).

Contradiction. A relationship in NLI where the second sentence is false if the first sentence is true, indicating a logical conflict.

Davao City Ordinances. These are written local laws created by the Davao City government that are intended to be permanent and enforceable.

Entailment. A relationship in NLI where the first sentence (premise) proves that the second sentence (hypothesis) is true.

Ex-Ante. Derived from Latin meaning "before the event". Operationally, this refers to the evaluation or checking of a proposed city ordinance before it is officially signed into law.

Information Retrieval (IR). The process of searching for and finding relevant documents (like specific laws) within a massive collection of text.

Local Government Code of 1991 (RA 7160). The national law that gives local governments the power to create their own ordinances for the "General Welfare" of their citizens.

Magtajas v. Pryce Doctrine. A legal rule from the Supreme Court stating that local ordinances are inferior to national laws and cannot go against any Act of Congress.

Natural Language Inference (NLI). A task where the computer compares two sentences—a "premise" and a "hypothesis"—to see if they agree or disagree with each other.

Neutral. A relationship in NLI where the two sentences are unrelated and do not affect each other's truth.

OCR Noise. Text errors (like misspellings or broken words) that happen when physical paper documents are scanned into digital files.

Sangguniang Panlungsod (SP). The local legislative body or "City Council" of Davao City, responsible for creating and approving local laws.

Semantic Conflict. A logical clash where two different laws give opposite instructions. For example, a conflict occurs if one law permits an activity while another strictly prohibits it.

Chapter 2

Literature Review

2.1 Legal Informatics Conflict Detection

For a legal system to work properly, its laws must be consistent. This means that citizens and the government should be able to understand their rights and duties without running into laws that give opposite instructions [36]. Semantic Conflict Detection—also called the detection of statutory or normative conflicts in recent studies—is the computer's task of finding these logical clashes between different legal rules [75, 2]. Specifically, it identifies cases where two or more valid laws give opposite orders to the same person under the same situation, such as when one law says you "must" do an act while another law "forbids" it [2, 58]. It is also important to distinguish between a new law that is meant to change an old one (a valid amendment) and a law that accidentally goes against an old one without saying so (a legal error) [53]. In the world of computer science and law, the move from manual reading to automated conflict detection is necessary because there are simply too many laws for humans to track. Ruhl and Katz [109] found that the number of new regulations has grown so fast that human experts can no longer keep up with all of them. As Shao et al. [116] points out, while lawyers are good at resolving known conflicts, it is physically impossible for them to cross-reference thousands of old laws at the same time to find hidden contradictions. Therefore, Semantic Conflict Detection acts like a "health check" for laws, using math to map out where different rules overlap so that problems can be fixed before they lead to court cases or government confusion. To understand how a computer can detect these conflicts, we must look at the factors that make it difficult. This section will examine the challenges of "legal inflation" (the growing number of laws), the limits of manual auditing, and the confusion caused by legal language. This section will also look at conflict detection research within the Philippines, specifically focusing on the hurdles found in local government archives.

2.1.1 The Unstructured Text Problem in Legal Corpora

Dealing with "unstructured" legal text—meaning text that isn't organized into a neat database—is not just a data problem; it is a failure in how laws are maintained. To build a system that works, it must be able to handle three specific issues: the massive amount of text, the slow pace of manual review, and the confusing language used in drafting. These three factors are why the field is moving away from human oversight and toward deep-learning tools for searching and checking laws.

2.1.1.1 Legal Inflation

We refer to ""Legal inflation" as to how the number of laws and their connections keep growing. Katz et al. argue that the sheer volume of laws has made it too difficult for humans to check for consistency manually [64]. As new rules are made to keep up with technology, they create "regulatory debt," where new laws might conflict with older ones without anyone noticing. This is a big problem for local governments, because unlike national laws, local ordinances are often not organized or scanned properly. Dias et al. [32] notes that turning these laws into machine-readable data is hard because they are often scattered across different files, from formal resolutions to quick executive orders.

The "search space" for finding conflicts is therefore a messy collection of different texts rather than a neat database. Zhong et al. [151] describe this as a "dynamic consistency problem" where facts and rules are always changing. Because this data is not organized, government agencies cannot be proactive. They only realize there is a conflict after something goes wrong. This is why the field has moved from old AI systems that use simple rules to new "neural models" that can learn from data. These systems are needed to process huge collections of laws that are too big for the human brain to handle [22].

2.1.1.2 The Manual Review Bottleneck

Scanning documents into a computer is no longer the main challenge; the real "bottleneck" is now checking if those documents agree with each other. As legislative bodies pass laws faster, they create a massive pile of unorganized data. The main difficulty is telling the difference between a "good" overlap (where a law is meant to cover multiple areas) and a "bad" conflict (where a law is written poorly and contradicts another). Recent surveys show that standard search engines usually fail to understand these nuances [8]. As a result, many conflicts stay hidden until someone gets sued. This moves the burden of finding conflicts from the people writing the laws to the judges in court [127].

2.1.1.3 Semantic Ambiguity

Old computer search systems look for exact word matches. However, this fails to catch the real meaning of human language. Legal language is like a specialized code. Words like "shall," "must," or "may" have very strict meanings that keyword searches often mix up. Chalkidis et al. [22] show that this "ambiguity" makes exact-match techniques fail at complex legal tasks. For example, a law banning "motor vehicles" might conflict with a law allowing "electric scooters." A keyword search for "vehicle" will not find the problem if the words do not match exactly.

Legal consistency often depends on "implied" rules—things the law suggests but doesn't say directly. Rigano et al. [107] argue that finding these contradictions requires a system that can understand logic, not just compare strings of text. "Transformer" AI models are an example of a solution that can solve this gap by creating mathematical representations (vectors) that capture deep meaning [150], which shall be discussed in later sections. However, there is a trade-off. Sovrano et al. [119] argue that while these models are very accurate, their "black box" nature is a problem because they don't explain *why* they found a conflict. In a legal setting, the reasoning is just as important as the detection [115].

2.1.2 Semantic Conflict Detection in the Local Context

While researchers around the world are building conflict detection tools using benchmarks like LexGLUE, these tools are still in the early stages in less prominent countries like the Philippines [22]. Since the Local Government Code of 1991, cities like Davao have created thousands of ordinances that are not well-organized [43]. Although the math for detecting conflicts exists in general AI research, it has rarely been applied to Philippine laws.

2.1.2.1 Legal-Related Semantic Conflict Detection in the Philippines

When Philippine AI research does focus on law, it is usually about organizing data rather than deep reasoning. Early work by Virtucio et al. [135] used basic classifiers to predict Supreme Court decisions, while Sagum et al. [111] used AI just to summarize national court rulings.

More recently, studies have started using "Transformer" models, a method that shall be discussed in later sections, but they are still used mostly for search. Roque et al. showed in 2024 [108] that "dense embedding" models are much better than keyword searches at finding specific tax laws. However, these embedding models are basically just advanced search engines. They can find two laws about "business taxes" because they are topically similar, but they lack the logic layers needed to tell if one law actually contradicts the other.

Looking at these trends, there is a clear gap in the research. The Philippine AI community has successfully used computers to categorize, summarize, and search national laws. They have also proven that NLI models can understand logical contradictions in local dialects for fact-checking [135, 111, 108, 94]. However, actually using these advanced NLI models to find conflicting rules in the messy, unorganized ordinances of a local city like Davao remains almost entirely unexplored.

2.1.2.2 The Landscape of Logic and Conflict Detection in the Philippines

In the Philippine AI community, the tools needed for conflict detection, specifically Natural Language Inference (NLI), are already being used, but mostly for other tasks. Recent local studies have used NLI to find logical contradictions in things like online news and misinformation. For example, Nuñez et al. proved that these AI models can handle the logic needed for Tagalog and "Taglish" (a mix of Tagalog and English) [94].

However, when researchers try to analyze local government records, they usually go back to traditional methods. Most document analysis in the Philippines still relies on "Boolean" search (keyword matching) and simple rule-based AI. As recent Philippine studies show, these old systems are "lexically fragile" [88]. They fail if two laws use different words for the same thing (like "fireworks" vs. "pyrotechnics") and they break down when there are typos or scanning errors (OCR noise) in the digital files. Trying to use manual rules to find clashes in thousands of messy city ordinances is simply too hard for a computer or a person to handle alone.

2.1.2.3 Gaps in Local Legislative Conflict Detection

While the Philippine AI community has successfully automated the search and summarization of national laws, a significant gap remains in using deep logic models (NLI) to audit local municipal ordinances. Research suggests this gap exists because local legislation is far more difficult for a computer to process than national statutes. This section outlines the specific reasons why this area of research remains largely unexplored.

Dataset Gap. The primary barrier to research is the lack of "ground truth" datasets for Philippine ordinances. Unlike national laws, which are well-documented, local ordinances are considered "low-resource" data [103]. Standard AI models are trained on massive Western datasets like LexGLUE, but there are no equivalent datasets that provide annotated pairs of conflicting Philippine ordinances to train an NLI model [6, 90]. Without a standardized benchmark to measure progress, local researchers have found it difficult to build systems that move beyond simple keyword matching to actual logical reasoning [6].

Resource Gap. Deductive logic models like Cross-Encoders require significant computing power that many Local Government Units (LGUs) simply do not have. Research on Philippine public administration shows that AI adoption is still in its early stages due to limited funding, a lack of skilled IT personnel, and outdated computer hardware [37]. Studies have shown that in some municipal councils, only 20 percent of legislative staff have received formal ICT training, and most still rely heavily on printed documents rather than structured databases [37, 6]. This lack of digital infrastructure makes it difficult for researchers to justify developing complex AI models that would be too "heavy" to run on actual LGU workstations [55, 6].

Linguistic Gap. Local ordinances are linguistically more complex than national laws because they frequently blend English with regional dialects like Cebuano and administrative "Taglish" [33]. Most Natural Language Processing (NLP) tools are designed for standard English and often break down when they encounter local terms or Spanish legal loanwords common in the Philippines [84]. Research on Philippine social media AI has proven that NLI can handle local dialects for simple fact-checking, but applying this same logic to the dense, multi-generational language of local ordinances is a much harder task that requires specialized domain knowledge that has not yet been fully researched [24, 46].

Trust Gap. There is also a trust gap within the legal profession. As noted by Supreme Court Senior Associate Justice Marvic Leonen, the legal community has not yet fully embraced AI because current models are often "black boxes" that do not explain their reasoning [73]. Lawyers are cautious about using automated tools because inaccuracies could lead to legal errors or the passage of void laws [3]. Consequently, research has focused more on simple organization and retrieval tasks—where the computer acts as a simple search engine—rather than deep reasoning tasks where the computer makes a judgment about a legal conflict [9].

2.2 The Philippine Legislative Domain

The Philippine legal system is a unitary state. This means that Local Government Units (LGUs) only have powers that are given or delegated to them by the national government [43]. This setup creates a strict top-down hierarchy that limits how much freedom cities like Davao City have when making laws. Any computer system built to check these laws must understand this specific legal environment. This section explains the barriers in Philippine municipal law. First, it looks at the strict rules that make local ordinances follow national laws. Second, it examines the difficulties and language issues in the local drafting process. Finally, it reviews the current methods used to check for consistency, which shows why we need to move toward automated auditing.

2.2.1 Hierarchical Constraints

The legal system in the Philippines depends on the rule that national laws always override local ones. To understand the limits of writing local laws, we must look at the specific rules that put ordinances below national statutes. This part describes the centralized framework of the government, the powers given by the Local Government Code, and the tests used by the Supreme Court to cancel local laws that conflict with national ones.

Unitary Principle in the Philippines. Research on Philippine political law shows a rigid pyramid of power: the 1987 Constitution is at the very top, followed by national laws (Republic Acts) passed by Congress, international treaties, executive orders, and finally, local ordinances [43]. Under this setup, no LGU can ignore or contradict an Act of Congress. The rule of "national preemption" is a major constraint; when a national law already covers an entire subject—such as firearms or immigration—local governments are not allowed to make different rules [43].

The Local Government Code of 1991. The main law for LGUs is the Local Government Code (LGC) of 1991. Section 16, known as the General Welfare Clause, gives LGUs the power to create ordinances for the safety and success of their citizens, but only if these rules stay "within the framework of the law" [43]. Since Davao City is a "Highly Urbanized City" (HUC), it is independent of any province and its ordinances are not checked by a provincial board. This makes the city more responsible for checking its own laws for conflicts before they are passed [43].

Magtajas vs. Pryce Doctrine. Perhaps the most important test for whether an ordinance is valid comes from the Supreme Court case *Magtajas v. Pryce Properties Corporation, Inc.* The Court listed six requirements that a local law must meet to be valid [123]:

- It must not contravene the constitution or any statute.
- It must not be unfair or oppressive.

- It must not be partial or discriminatory.
- It must not prohibit but may regulate trade.
- It must be general and match public policy.
- It must not be unreasonable.

Local ordinances are considered lower in rank than state laws. Therefore, LGUs cannot try to regulate activities that the national government has already permitted or regulated [43].

2.2.2 Procedural and Linguistic Friction

Besides the hierarchy of laws, the actual way local laws are written and passed creates a lot of inconsistencies in language and format. National laws are usually written by specialized committees using standard legal terms. However, local ordinances go through a localized process that often breaks up formal language. To detect conflicts accurately, the computer system must be adjusted to handle the "noise" and confusion caused by using two languages in the Davao City Council (Sangguniang Panlungsod).

Ordinance Creation Process. In Davao City, the life of a proposed law is guided by the Internal Rules of Procedure, found in Resolution No. 001-22 [112]. The drafting process is decentralized, meaning different council members or their staff write their own laws rather than having one main office do all the writing [43, 112]. Because of this, the style and words used across different ordinances can be very inconsistent. For example, a transportation law might use completely different terms than an environmental law, making it hard for computer models to see how they relate.

Textual Fragmentation and the Three-Reading Rule. According to the law, no ordinance can be passed unless it goes through three separate readings [43, 112]. This process is democratic, but it often breaks the text apart. During the First Reading, the draft is sent to a committee. The Second Reading is where the most intense checking happens: the text is put into a period of interpolation, debates, and amendments [112]. This constant changing of the text can lead to disjointed rules. For instance, a new exception added in one section might accidentally contradict a general rule in another section, which may contain a hidden conflict that stays buried in the text after final approval on the Third Reading.

Bilingual Mandates and the Risks of Translation. A unique challenge within the Philippines is the legal requirement to make laws accessible in two languages. Under Section 469 of the Local Government Code, the Secretary must translate all approved ordinances into the local dialect (e.g. Cebuano for Davao City) [43, 112, 33]. However, this creates a legal risk. The Administrative Code of 1987 (Executive Order No. 292) says that if the English and local versions disagree, the English version is the one that counts [105]. Consequently, if a translation error happens, a citizen following the local language could be punished for breaking the English version of the law. This relates to the Supreme Court ruling in *Tañada v. Tuvera*, which says laws are void without proper public notice [121]. A mistranslated law is essentially "false notice." Despite new efforts like the *Batas sa Sariling Wika Act* to mandate more regional translations [99], there is still no technical system to check if these translations are actually correct. As researchers have noted, the small details in local languages are often ignored, leading to "semantic drifts" where the original legal meaning in English is lost in regional translation [84].

2.2.3 Ex-Ante in Legislation

The term "ex-ante" comes from Latin and means "before the event." In the context of lawmaking, it refers to the process of evaluating a proposed law or regulation *before* it is officially enacted [35]. This part looks at how the Philippines uses ex-ante reviews to prevent bad laws from being passed and how this study fits into that process.

Ex-Ante in the Philippine Context. In the Philippines, ex-ante review has become more important since the passage of Republic Act No. 11032, or the "Ease of Doing Business and Efficient Government Service Delivery Act of 2018" [44]. Under this law, the Anti-Red Tape Authority (ARTA) and the Department of the Interior and Local Government (DILG) mandates that government agencies and local government units (LGUs) conduct a Regulatory Impact Assessment (RIA) for any proposed regulation that might significantly affect the public [44]. The goal is to ensure that new rules have a solid legal basis and to avoid creating overlapping or conflicting regulations that could make it harder for citizens and businesses to operate.

Ex-Ante vs Ex-Post. The main difference between the two is timing and purpose. Ex-ante evaluation is predictive; it looks forward to stop problems before they happen. It acts as a "first line of defense" against poor lawmaking [35]. In contrast, ex-post evaluation happens *after* the law is already active, focusing on how the law has actually performed or whether it has caused damage [129]. While ex-post reviews are useful for fixing existing mistakes, they are often difficult because it is much harder to revoke a law once it is already being enforced [129]. Ex-ante detection is considered more efficient because it saves the government from the cost and trouble of litigation later on.

Conflict Detection in Local Ex-Ante Legislation. Using automated tools for the ex-ante Conflict Detection can help the local legislative branch identify "normative collisions" during the drafting phase. Because legislative bodies now produce a massive amount of text, manual review often fails to catch hidden clashes between a new ordinance and older statutes [64]. By treating Conflict Detection as a diagnostic triage tool, the system can flag potential contradictions while the ordinance is still just a draft. This ensures that the final law is consistent with the national legal hierarchy, protecting the city from passing ordinances that are legally void from the start [123].

2.2.4 Consistency Checking in Legislative Processes.

Because local governments are lower in rank and their drafting process is fragmented, checking for consistency is a critical duty. However, checking ordinances for conflicts is not as simple as matching titles, as it requires understanding complex legal meaning and identifying subtle oversteps in authority. This part examines current methods of consistency checking, highlighting how confusing language and the reliance on simple tracking systems slow down the legislative review process.

Legalese Linguistic Ambiguity. Discourse analysis of Philippine laws reveals many ambiguities in how words and sentences are used [122]. Specifically, the word "shall" is often confusing because it can be interpreted as either a mandatory command or a simple suggestion depending on the sentence. Studies show that these language issues make it hard to interpret laws correctly, even when they follow standard formats [12].

Legislative Gaps and Implied Preemption. The Philippine legal framework recognizes that conflicts between laws are rarely obvious [28]. A local law almost never says, "This law contradicts a national law." Instead, conflicts often appear as "implied preemption." This happens when a national law already covers a subject, but a local government tries to add extra hurdles—like requiring a local fee to do something already allowed by the state [122]. In these cases, the Supreme Court rules that the local ordinance is unlawfully trying to "amend" a national law under the guise of local regulation [122]. For a human reviewer or a basic keyword search, this is very hard to find because the laws might use different words. The conflict is in the actual effect of the rules, which requires deeper semantic checking to identify.

Local Context. The Davao City Sangguniang Panlungsod passed nearly 400 ordinances and over 2,000 resolutions in a single three-year term [97]. To manage this volume, the city uses digital platforms: the Resolution Ordinance Tracking System (ROTS) is used for encoding and routing documents, while the Legislative Information Support System Program (LISSP) lets council members track items and read ordinances on their devices [24]. However, while these systems help with logistics, the actual checking of content for "dueling provisions" is still done manually. Furthermore, scanned PDFs often have OCR noise, which makes keyword searches fail [78].

2.3 Philippine Digital Legislative Archives

Building a functional Semantic Conflict Detection system requires access to comprehensive and machine-readable legal corpora. In the Philippines, the digital landscape of legislative data is highly fragmented, split between national statutory repositories and localized municipal archives. This structural division dictates how the initial data gathering phase must be engineered to capture all relevant textual evidence. Before any deep logical inference can occur, the architecture must first secure reliable data streams from these disparate legal environments. Consequently, understanding the specific characteristics of these sources informs the design of the retrieval and preprocessing pipelines.

2.3.1 National Sources

National legal repositories represent the authoritative apex of the Philippine legislative hierarchy, housing laws that universally preempt local ordinances. These databases are maintained by centralized government agencies and contain highly formalized texts such as Republic Acts, Commonwealth Acts, and Batas Pambansa. They serve as the primary knowledge base for the system's external legal context, providing the baseline rules against which municipal drafts are evaluated. Because national legislation undergoes a rigorous, standardized drafting process, the vocabulary within these repositories is generally consistent and adheres to strict legal English paradigms. Accessing these national databases provides the computational framework with the necessary macroscopic view of state-level mandates and prohibitions.

Official Gazette. The Official Gazette serves as the mandated repository for Philippine statutes, rendering it the official journal of the country for legislative text [45]. It acts as the primary publisher for all newly enacted national laws, executive orders, and administrative issuances, as mandated by decree of Republic Act No. 453 [98] and Commonwealth Act No. 638 [89]. Because it represents the final, legally binding version of a statute, any computational system evaluating municipal preemption must anchor its baseline rules to this repository. This archive effectively acts as the definitive historical ledger for state-level mandates, preserving the exact phrasing approved by the legislative and executive branches. Consequently, it remains the most critical target for establishing the system's foundational legal knowledge base.

Open Congress API. To programmatically map the complex hierarchy of national laws, developers utilize the Open Congress API provided by the House of Representatives' Legislative Information System (LEGIS) to access structured, machine-readable metadata regarding legislative routing and amendments [54]. This interface efficiently tracks a statute's formal lifecycle, charting its progression from initial committee filing to final executive approval. By providing standardized output formats, the API allows computational pipelines to establish relational dependencies between different legislative acts without manually parsing unstructured text.

The LawPhil Project. The Lawphil Project functions as a massive, privately maintained database of statutory records and jurisprudential decisions [7]. Unlike official government channels, this repository provides a comprehensive, pre-digitized plaintext corpus of historical and contemporary Philippine law. This accessibility makes it mathematically ideal for the unsupervised pre-training of foundational language models tailored to the local legal domain. By ingesting this vast corpus of clean text, NLP architectures can learn the highly specific syntactic structures and deontic operators inherent in Philippine jurisprudence. Utilizing this database effectively establishes the core vocabulary required for advanced Transformer models to process specialized statutory drafting [150].

2.3.2 Local Sources

In contrast to the centralized national databases, local legal repositories are maintained independently by individual Local Government Units (LGUs). These archives house municipal ordinances, resolutions, and executive orders that govern specific geographic jurisdictions. The legislative output at this tier is entirely decentralized, resulting in highly variable archiving practices depending on the technological capacity of the specific city council. These local databases form the immediate target corpus for the conflict detection system, containing the actual regulatory constraints imposed on city residents. Consequently, they represent a high-volume, dynamic data environment that continuously expands as the local council holds regular drafting sessions.

Davao City Sangguniang Panlungsod. The locality's primary corpus consists of the Davao City Sangguniang Panlungsod (City Council) website, which serves as the official digital repository for the city council's legislative outputs [25]. This database acts as the primary historical record for all municipal regulations, resolutions, and zoning policies enacted within the city's jurisdiction. It represents a dynamic and continuously expanding text environment that directly reflects the decentralized nature of local lawmaking. The texts housed here encapsulate the specific regulatory constraints and policy directions of the city government, providing the exact ordinance text required to evaluate new drafts for logical consistency against the existing legislative framework.

Legislative Information Support System. To manage the sheer volume of local legislation, the city introduced the Legislative Information Support System Program (LISSP) as its primary administrative backbone [24]. This platform serves as a digital tracking system designed to maintain metadata regarding ordinance routing, committee assignments, and plenary approval statuses. According to official reports, the system is openly available to the public, allowing citizens and stakeholders to monitor ordinances and resolutions in real-time [86]. By digitizing the legislative workflow, LISSP effectively functions as a centralized database of ordinances, providing the structured tracking and contextual data necessary for evaluating the lifecycle of local laws. This public transparency and administrative organization address the interoperability challenges identified in recent regional e-governance assessments [3].

2.3.3 Data Extraction Challenges

While the aforementioned repositories contain the necessary legal knowledge, extracting the needed data for computational analysis presents severe technical bottlenecks. Scraping and formatting these diverse sources exposes the pipeline to significant data degradation and structural inconsistencies.

Optical Character Recognition. A preliminary survey of the Official Gazette and the Davao City SP archive reveals a significant barrier to automated auditing: the lack of native, machine-readable text. Both repositories predominantly host scanned, image-based PDFs rather than "searchable" or "selectable" digital documents. During manual inspection, these files were observed to contain varying degrees of visual noise, including slanted scans, faded ink, and compression artifacts. As Lopresti [78] documents, such "noisy" visual data often results in severe Optical Character Recognition (OCR) degradation, where standard extraction tools produce broken alphanumeric strings and nonsensical tokens. These observed irregularities suggest that a direct pipeline would suffer from semantic fragmentation, requiring an additional, high-precision OCR layer and extensive text-cleaning heuristics to ensure the data is viable for program analysis.

Archival Fragmentation. Another critical challenge is archival fragmentation, a state where repositories suffer from systemic gaps in their digital records. This is a significant problem for high-fidelity auditing because the absence of even a single related ordinance can lead to "false negatives" in Conflict Detection, where a system declares a draft valid simply because it cannot see the contradicting record. One primary example is the Davao City Sangguniang Panlungsod public archive, which features intermittently missing ordinance files and unorganized chronological listings that actively hinder systematic data collection [113]. Furthermore, even relatively complete national databases like the Lawphil Project can lack specific Republic Acts or historical amendments. As Westermann et al. [49] observe, legal databases are frequently "unstructured" and "fragmented," forcing developers to move beyond simple scraping toward complex data engineering to ensure a "complete" dataset. This requires developing specialized, fault-tolerant scripts to account for missing files and broken links, significantly complicating the data-gathering phase required to build a reliable local knowledge base.

Database Accessibility. A significant challenge in data acquisition is the discrepancy between the theoretical and actual accessibility of local government databases. One notable example of this is with LISSP, because while the Program is promoted in institutional reports as a tool for public transparency [24, 86], empirical verification reveals a strict authentication barrier requiring login credentials for entry. Upon technical inspection by the proponents, the LISSP portal remains computationally isolated behind a login interface, effectively restricting access to internal government personnel and preventing programmatic extraction by public scrapers as well as access to the general public. This "gatekept" nature of the database was further confirmed through direct correspondence with system administrators by the proponents, where it was discovered that the archive is not open in a digital sense; access for this study was only secured through formal requests and manual persuasion via email. As Alfiani et al. [3] observe in similar administrative systems, such isolated data often exists as "digitally orphaned" artifacts, where metadata about an ordinance is entirely disconnected from its full-text searchable content. This structural disconnection forces an automated consistency-checking pipeline to bypass the LISSP's restrictive front-end and instead engineer custom preprocessing logic that attempts to mathematically re-bind any available public metadata to the continuous vector representations of the offline, full-text documents gathered from alternative, non-authenticated archives.

Textual Scarcity. Perhaps the most significant challenge in developing an automated legal auditing system is the pervasive systematic textual scarcity within the Philippine legal informatics landscape. Despite the existence of authoritative sources like the Official Gazette and the Davao City SP website, their web architectures remain inherently hostile to modern programmatic extraction, characterized by unindexed entries and "deep-linked" PDFs that cause automated scrapers to fail. Even community-driven initiatives or secondary repositories do not offer a single, straightforward solution. For instance, while the Open Congress API provides structured metadata for legislative routing, it lacks the full-text bodies of the laws themselves. Conversely, while the Lawphil Project offers a more consistent link format and a vast plaintext corpus, it is not an exhaustive archive and contains occasional missing Republic Acts or amendments. As Shao et al. [116] observe, such fragmentation prevents the construction of a unified, high-fidelity knowledge base from a single origin. Consequently, researchers are forced into a series of technical "roundabouts" and compromises—engineering a patchwork pipeline that pulls metadata from one source and text from another—to circumvent the absence of a centralized, machine-readable repository for both national and local legislation.

2.4 Exact and Symbolic Methods

Before the adoption of neural networks, computational legal analysis relied heavily on deterministic systems to process text. Deterministic systems operate on absolute rules, meaning that a specific input will invariably produce the exact same output without any probabilistic guessing. Traditional methods for searching and evaluating legal information assumed that legislative ideas could be cleanly mapped either to specific vocabulary strings or to rigid, hand-coded logic formulas. While highly transparent and easy to debug, these foundational approaches frequently fail when forced to detect hidden conflicts in complex, unstructured legal texts. The core issue is that human language is inherently messy, and rigid systems break down when faced with the ambiguous phrasing typical of municipal ordinances.

2.4.1 Boolean Information Retrieval

To understand why traditional legal databases struggle with semantic conflict detection, we first need to look at how they fundamentally process text at the machine level. Boolean Information Retrieval (BIR) represents the foundational search architecture for almost all commercial law databases currently in use. Instead of reading a document like a human, BIR treats every file simply as a "bag of words," stripping away all grammar and sentence structure. It relies on an inverted index, which is essentially a massive lookup table that tracks exactly which specific words appear in which files across the entire database. Because it only registers the presence or absence of characters, the system is completely blind to the actual context or legal intent behind those words.

At its core, the retrieval mechanism in BIR is strictly based on set theory and formal Boolean logic. A user constructs a query using logical operators like `AND` ($\wedge$), `OR` ($\vee$), and `NOT` ($\neg$) to create intersecting sets of documents. For example, a search for a specific ordinance regarding explosives might be formulated mathematically as:

$$Q = (\text{fireworks} \vee \text{pyrotechnics}) \wedge \neg\text{exempt}$$

The system then retrieves documents by calculating the exact mathematical intersection of these sets, pulling only the files that satisfy every condition simultaneously. If a document contains the specific string of characters requested, it is retrieved and scored as a 1; if it misses even a single required term, it is completely ignored and scored as a 0. It evaluates the raw presence of strings, entirely ignoring the conceptual meaning or legal nuance behind them [14].

2.4.1.1 Challenges with BIR

The primary limitation of BIR in legal auditing is its absolute, uncompromising reliance on exact string matching. This rigidity creates a severe "vocabulary mismatch" problem when processing natural language. Human language naturally features synonymy, where entirely different words possess the exact same meaning, and polysemy, where the identical word takes on drastically different meanings depending on the context. In statutory drafting, one author might use "prohibited" while another uses "unlawful," referring to the exact same legal concept. Because BIR operates on character-level matching rather than conceptual mapping, it fundamentally cannot connect these two terms, leading to massive blind spots in the search results.

A landmark empirical evaluation of these systems demonstrated this vocabulary gap in a highly practical setting. Legal researchers conducting queries using standard BIR systems confidently estimated they were finding over 75 percent of the relevant documents for a given case. They experienced an illusion of effectiveness because the documents they did retrieve looked highly accurate, masking the massive volume of relevant files they missed. In reality, the system retrieved less than 20 percent of the actual relevant files contained within the database [14]. The researchers missed the remaining 80 percent simply because the authors of those hidden documents used slightly different, completely valid phrasing that didn't perfectly align with the strict search query [131].

In a local government context like Davao City, this specific margin of error is administratively unacceptable. Missing even one existing ordinance during an ex-ante review due to a simple vocabulary mismatch could directly result in the passage of an invalid, conflicting law. Furthermore, legislative vocabulary inherently evolves over time as drafting conventions change. An ordinance passed in the 1990s and a modern draft written in 2026 might strictly regulate the exact same municipal subject but share absolutely zero common keywords. Recent studies emphasize that resolving this requires the implementation of "Transformer" models, which abandon exact matching entirely in favor of continuous vector spaces that recognize when different vocabulary points to the same underlying concept [67].

2.4.2 Symbolic AI

Because BIR fundamentally fails to capture underlying meaning, early artificial intelligence researchers attempted to automate legal reasoning by directly modeling the rules of law themselves. This approach, widely known as Symbolic AI or "Good Old-Fashioned AI" (GOFAI), operates on the premise that human cognition can be replicated by manipulating formal symbols [133]. It moves beyond simple word matching by teaching computers actual legal logic through explicit, human-coded "IF-THEN" formulas. Instead of searching a database for raw text strings, Symbolic AI requires human engineers to manually translate the constraints of a statute into formal logic structures. This theoretically allows the computer to process the actual regulatory mechanics of a law rather than just reading its surface vocabulary.

To execute this logic, these systems specifically utilize propositional or first-order logic combined with deontic operators. Deontic logic is a highly specialized branch of mathematical logic that deals specifically with obligations, permissions, and prohibitions. For example, a system might represent a local zoning prohibition mathematically as:

$$\forall x(\text{Business}(x) \land \text{ResidentialZone}(x) \to \text{Prohibited}(x))$$

In this formula, the universal quantifier ($\forall$) asserts that for any entity x, if it is classified as a business ($\land$) and located in a residential zone, it is strictly prohibited ($\to$). By converting ordinances into these rigid formulas, a system can theoretically use deductive reasoning to spot a conflict. If Rule A dictates $\mathcal{O}(X)$ (an absolute obligation to do X) and Rule B dictates $\mathcal{F}(X)$ (an absolute prohibition against doing X), the system successfully flags a formal logical contradiction ($\perp$) [10].

2.4.2.1 Challenges with Symbolic AI

While Symbolic AI provides perfectly transparent and mathematically sound reasoning, its absolute reliance on manual programming makes it practically impossible to scale for real-world governance. The primary architectural barrier is known as the "knowledge acquisition bottleneck." Human domain experts and software engineers must manually read, interpret, and translate thousands of unstructured city ordinances into rigid computer rules. This manual translation process is unacceptably slow, extremely expensive, and highly prone to human error during the coding phase [81]. Furthermore, every time the city council amends an existing law, the underlying symbolic formula must be manually rewritten, making the system incredibly difficult to maintain over time.

Beyond the logistical nightmare of manual coding, the rigid 1-and-0 architecture of Symbolic AI fundamentally clashes with the "open texture" of natural legal language. Laws frequently rely on subjective, open-ended standards—such as requiring a citizen to exercise "reasonable care" or negotiate in "good faith." These inherently subjective human concepts cannot be easily defined within a strict, binary rule-based formula because their definitions change depending on the specific context [9]. While these symbolic systems are highly precise in tightly controlled, hypothetical environments, they immediately shatter when exposed to the real world. Ultimately, they lack the computational flexibility required to handle the thousands of messy, unorganized, and ambiguously worded documents actually found in local government archives [151].

2.5 Graph-Based Architectures

Because exact word matching can be unreliable, some researchers use "graph-based" systems. Instead of looking at laws as just lists of words, these systems treat them like points in a large web. By looking at how laws quote each other and relate to the same topics, the computer can better understand their meaning.

2.5.1 Legal Knowledge Graphs and Graph Neural Networks

Legal Knowledge Graphs (LKGs) are a way to model how different laws depend on each other. Unlike a simple list, these graphs can use Graph Neural Networks (GNNs) to find hidden links and clashes. Researchers have created frameworks like GNN-ERE to automatically build these legal webs from messy text, which works much better than just searching for keywords [83]. Studies also show that these networks can find relevant court cases even if they do not use the exact same words [50].

The advantage of GNNs is that they can share information across the entire web structure. By treating laws as points in a network, the computer can learn the "purpose" of a law—for example, telling the difference between a rule and a procedure [13]. These systems can also perform "multi-hop reasoning." This means they can find "indirect conflicts" where two laws do not quote each other directly but still clash because of their relationship with a third law [145]. This is a big step up from old search engines that can only look at one thing at a time.

2.5.1.1 Challenges in GNNs

While Graph Neural Networks are powerful, they only work if the web of connections is high-quality. A GNN needs a clearly built index or list of links to follow.

Dependency on Clear Connections. GNNs are limited because they need clear links between laws to "see" the meaning. As researchers have pointed out, creating these links by hand is very slow and expensive [57]. In a city with thousands of ordinances, it is almost impossible for people to keep this list of links up to date. If the system fails to find a link between two conflicting laws at the start, the GNN will be "blind" to that problem.

Disconnection in Digital Archives. Even though systems like the Legislative Information Support System Program (LISSP) have helped move Davao City's archives online, a gap remains for graph-based AI. Most digital repositories in local government are designed for storing and viewing files rather than linking ideas together. Recent studies on Philippine digital governance show that local ordinances often act as "digitally orphaned" texts; they are available online but lack the machine-readable links needed for an AI to jump from one law to another [3]. This makes the digital archive a collection of separate records rather than a connected network, making it hard for GNNs to see how they overlap.

Weakness Against Messy Text and OCR Noise. Building these legal webs automatically is also very difficult because of how local government data looks. AI frameworks usually assume the text is "clean" and easy to read [83]. However, scanned PDFs in Philippine archives often have "OCR noise" (typos from scanning), which confuses the computer [78]. When there are typos or local "Taglish" words, the computer creates broken or "hallucinated" links. This makes the whole web unreliable for finding actual legal clashes.

2.5.2 Abstract Meaning Representation

While GNNs look at links between documents, Abstract Meaning Representation (AMR) looks at the logic *inside* a single sentence. AMR is a graph system that focuses on the "who is doing what to whom" logic. Researchers argue that this provides a more stable way to understand laws than just looking at grammar [136]. By mapping out the logic, AMR can find conflicts even if laws use different wording (like "The city bans X" vs "X is banned by the city").

AMR is also useful for identifying which rules are "obligations" (must do) and which are "prohibitions" (cannot do) [125]. Unlike old keyword searches, AMR understands the role of each word. This helps find subtle clashes, such as a local law allowing "business use" when a national law forbids "non-residential activity." New legal datasets have proven that AMR is a good way to combine human-like logic with computer power [136].

2.5.2.1 Challenges in AMR

Even though AMR is very precise, using it for Philippine city ordinances is difficult. It relies on very strict rules that don't always match the "messy" way local laws are written.

Limited to Single Sentences. The biggest flaw of AMR is that it only looks at one sentence at a time [125]. However, legal conflicts in city ordinances rarely happen in the same sentence; they are usually found in different sections or paragraphs. For example, a ban might be in Section 2, while an exception is buried in Section 5. Because AMR cannot "connect the dots" between different sentences easily, it is often "blind" to these types of conflicts.

Difficulty with Complex Legal Language. AMR relies on a static or fixed dictionary of meanings to work. The long, "run-on" sentences often found in Philippine laws are too complex for most AMR tools. For example, in RA 7160 within Section 378, such sentence can be found as follows:

"When a loss of government property occurs while the same is in transit or is caused by fire, theft, force majeure, or other casualty, the officer accountable therefor or having custody thereof shall immediately notify the provincial or city auditor concerned within thirty (30) days from the date the loss occurred or for such longer period as the provincial, city or municipal auditor, as the case may be, may in the particular case allow, and he shall present his application for relief, with the available evidence in support thereof."

Such a sentence causes the AI often fails or produces the wrong logic map, as studies show that as sentences get longer and more confusing, AMR becomes much less accurate [126].

Problems with Local Words and Typographical Errors. AMR is also very sensitive to typographical errors and unknown words. When the computer runs into a scanning error from an old PDF, or a local word that isn't in its dictionary, it fails to map the logic correctly. Research shows that this "lexical noise" often results in broken logic maps that lose the actual meaning of the law [130]. Without a tool specifically made for Philippine Legal English, relying on AMR for Davao City's archives would be unreliable.

2.6 Deep Learning and Transformer Models

The application of Natural Language Processing (NLP) in the legal domain has shifted from rigid, rule-based systems and basic word-matching toward Deep Learning [65]. Legacy algorithms treat legal texts as simple character strings, which frequently fail to capture the dense and interconnected nature of statutory drafting. Deep Learning, conversely, utilizes neural networks to map out the actual semantic meaning of legal language. This allows systems to process complex syntax and perform the deductive reasoning required to spot conflicts. This section breaks down the Transformer architecture, explaining its underlying mechanics, computational challenges, and why it remains the most practical framework for auditing unstructured municipal archives.

2.6.1 Transformer Models and Self-Attention Mechanism

A major breakthrough was the Transformer architecture, introduced by Vaswani et al. [132] in 2017. Prior models, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, processed text sequentially. This step-by-step processing was computationally slow and made it difficult for the model to retain information from earlier in a long document. Transformers bypass this sequential bottleneck entirely by relying on a Self-Attention mechanism.

Self-Attention allows a model to analyze all words in a sequence simultaneously to determine their contextual relationships. This enables a specific term to connect with any other word in the document, regardless of the physical distance between them. Because it processes data in parallel, Transformers are significantly faster to train, directly enabling the creation of Large Language Models (LLMs). Modern iterations have introduced refinements like Disentangled Attention to better parse complex syntactic structures, alongside Windowed Attention to process long-context legal documents efficiently. However, while Generative LLMs like ChatGPT excel at text summarization, they are prone to 'hallucinations'—often inventing plausible but entirely fake legal citations [29]. To ensure legal validity, a system must be discriminative: specifically classifying whether statements exhibit a Conflict or Entailment, rather than simply generating new text.

2.6.2 Challenges in Transformer Models

Despite these architectural advantages, deploying Transformer models within the hardware limitations of Philippine Local Government Units (LGUs) introduces specific computational hurdles.

The Quadratic Bottleneck and Context Limits. The Transformers' core processing engine scales quadratically ($O(n^2)$), meaning it requires an exponentially larger amount of memory as the input text grows [132]. Because of this "quadratic bottleneck," standard models like BERT are generally capped at reading 512 tokens at a time [31]. Philippine ordinances are frequently verbose, often spreading rules and exceptions across multiple pages. To manage this, developers use a **Sliding Window** technique. This involves taking a fixed-size "window" (e.g., 512 words) and moving it across the document with a specific **Stride** [5]. For example, if the window moves 256 words at a time, the windows will overlap, ensuring that the end of one chunk is also the start of the next to try and keep context together [52]. However, this is like reading a long book through a small slot; if a general rule is on Page 1 and its specific exception is on Page 5, the AI can never see them at the same time [115]. This physically cuts the "semantic link" between related clauses, creating logical blind spots where the system might flag a conflict simply because it cannot see the full picture [115].

Morphological Noise and Domain Adaptation. Transformers require massive corpora of clean text for proper pre-training. While specialized models like Legal-BERT [20] perform well for US or European law, there is no equivalent foundation model trained specifically on Philippine local legislation. Applying Western-trained tokenizers to local government archives—which are saturated with uncorrected OCR errors, archaic colonial phrasing, and administrative 'Taglish'—causes severe tokenization fragmentation. The model fails to recognize these localized terms and shatters them into meaningless sub-tokens, effectively ruining its ability to comprehend the law.

Edge Deployment and Hardware Constraints. Executing deep logical checks across thousands of document pairs demands significant computing power, typically requiring expensive GPU resources [104]. Most municipal governments simply lack this caliber of server infrastructure. Consequently, deploying these heavy models on standard LGU workstations requires aggressive optimization and decoupled pipeline architectures to ensure the system does not freeze during inference.

2.6.3 Transformer Models as the Ideal Method

Notwithstanding these hardware limitations, Transformer models remain the most viable paradigm for Semantic Conflict Detection in local ordinances. General-domain language models typically struggle with legal text because common words frequently take on entirely different legal definitions. For instance, "party" refers to a litigant rather than a celebration, and "suit" signifies a legal action rather than clothing. Greco et al. [46] emphasize that this high degree of polysemy is exactly why domain-specific models are required. While older static systems assign a single, fixed meaning to a word [46], a Transformer dynamically adjusts its reading of a word based on the surrounding sentence context [20]. Chalkidis et al. [21] empirically showed that this contextual awareness is what allows a model to distinguish between amending a law and violating it—a nuance that simple keyword searches invariably miss.

The transition to Deep Learning necessitates explicitly moving away from legacy methodologies. Boolean keyword searches fundamentally fail because legislative drafting vocabulary constantly evolves over the decades. Conversely, manually mapping thousands of unlinked Republic Acts and local ordinances into deterministic graph structures creates an impossible administrative bottleneck. As Li et al. [74] observed, hand-crafting these legal knowledge graphs simply does not scale for volatile, unstructured municipal records. Ultimately, Transformers represent the ideal method: they possess the capacity to map the deep, latent semantic relationships in unstructured legal text without requiring an intractable amount of manual data entry.

2.7 Transformer Pipeline

The appeal of an end-to-end, single-stage neural architecture is its operational simplicity: funneling raw text directly into final outputs without intermediate steps. For general Natural Language Processing (NLP) tasks like sentiment analysis, a single Transformer model is usually enough. However, legal informatics, specifically Semantic Conflict Detection, requires a level of deductive precision that a single monolithic model structurally cannot provide. Forcing one architecture to act simultaneously as a high-volume search engine and a strict logical judge creates impossible mathematical trade-offs.

2.7.1 Challenges in Single-Stage Pipeline

Deploying a single Transformer model to scan decades of uncodified municipal ordinances introduces severe architectural friction. The system inevitably compromises either its recall (missing conflicting laws) or its precision (flagging false positives). This limitation changes depending on the specific type of single-stage model used.

Generative LLM Constraints. A common but flawed approach feeds a document pair directly into a Generative Large Language Model (Decoder-only architectures like GPT or LLaMA) and prompts it to declare a conflict. These models are auto-regressive text predictors, not discriminative logical classifiers. As Linna [77] notes, applying them to strict statutory auditing invites 'hallucinations'—the fabrication of plausible but legally fake citations. Furthermore, their massive parameter counts make offline Edge Deployment on standard Local Government Unit (LGU) hardware impossible without aggressive quantization. Dettmers et al. [30] demonstrated that this quantization process severely damages zero-shot reasoning fidelity in specialized domains.

Bi-Encoders and their Limitations. A Bi-Encoder, also known as a "Dual-Tower" or "Two-Tower" model, is an AI architecture designed for high-speed searching [104]. It works by processing the user's search query and the documents in the archive completely separately from each other. The model turns every ordinance into a mathematical coordinate (a vector) and stores them in a database [63]. Because it does not have to read the documents at the same time a person is searching, it can compare a query against 10,000 national laws in just a few milliseconds [23]. This makes it the perfect tool for the "Coarse" stage of the pipeline, where the goal is to narrow down the entire legal archive to just a few relevant provisions [40].

Current municipal search systems predominantly rely on Dense Retrievers (Bi-Encoders). While highly efficient for filtering millions of documents, Thakur et al. [124] explain that Bi-Encoders suffer from a severe 'bottleneck of compression' because they collapse the semantic meaning of an entire document into a single, fixed-size vector. This architecture optimizes for topical similarity rather than logical consistency. Consequently, a Bi-Encoder has a critical logical blindspot: it will retrieve an ordinance "allowing fireworks" and one "banning fireworks" and score them as highly relevant because their vocabulary distributions are nearly identical. Kim et al. [67] observed that Bi-Encoders fundamentally lack the computational architecture to detect the logical contradiction operating beneath the shared syntax.

Overcoming this blindspot requires a secondary phase using a Cross-Encoder architecture. Instead of independently compressing documents into isolated vectors, a Cross-Encoder computes token-level attention across both documents simultaneously ([CLS] Ordinance A [SEP] Ordinance B). This allows the model's attention heads to directly compare negators (e.g., "banning" vs. "allowing") in their immediate contexts. Nogueira and Cho [93] demonstrated that this cross-attention mechanism shifts the task from basic topical retrieval to rigorous Natural Language Inference (NLI), enabling the precise Semantic Conflict Detection required for legal texts.

Quadratic Memory Constraints If a single-stage Bi-Encoder lacks logical depth, the theoretical alternative is running a single-stage Cross-Encoder across the entire database. However, this is computationally impossible. Because the Self-Attention matrix scales quadratically ($O(n^2)$) with sequence length, Beltagy et al. [11] highlight that cross-encoding a target ordinance against 10,000 municipal laws simultaneously instantly exceeds the available VRAM on existing hardware. The model would be forced to chunk the database into tiny, fragmented Sliding Windows, severing the critical semantic links between general rules and remote exceptions.

2.7.2 Single-Stage vs. Dual-Stage Pipeline

Given these hardware and logical constraints, the literature discourages single-stage monoliths for legal analysis. Expanding to a dual-stage pipeline is not just an optional optimization, as it is a structural necessity for processing high-volume, unstructured text. Several critical considerations justify this architectural split:

Retrieve-Then-Entail Paradigm. Legal auditing requires a shift from topical similarity to logical inference. A Retrieve-Then-Entail pipeline serves as a Coarse-to-Fine guardrail: an efficient retriever isolates candidates, and a deep-reasoning NLI Cross-Encoder identifies specific policy violations. Williams et al. [143] indicate that this approach reduces manual review time by up to 70 percent while maintaining conceptual coherence.

Hardware Feasibility. A dual-stage approach segregates the computational heavy lifting. Stage one relies on Information Retrieval via pre-computed vector indexes that use minimal active memory. This offloads the intense $O(n^2)$ matrix multiplications of the Cross-Encoder exclusively to the top-K retrieved pairs. Khattab and Zaharia [66] demonstrated that this allows advanced NLP to run locally on standard servers without requiring enterprise GPU clusters.

Pragmatic Solution to Compression. Splitting the pipeline lets each model specialize. The first stage accepts the bottleneck of compression to maximize search speed. As Humeau et al. [56] observed, the second stage avoids compression entirely, reading raw, uncompressed tokens across both texts to achieve high-precision logic.

2.7.3 Competition on Legal Information Extraction and Entailment

The necessity of architectural decoupling is highlighted by over a decade of results from the Competition on Legal Information Extraction and Entailment (COLIEE), moving beyond mere theoretical preference. As Goebel et al. [40] detail, COLIEE serves as the premier international proving ground for legal NLP, challenging research teams to solve complex jurisprudential problems using standardized datasets. Crucially, the competition structurally enforces the separation of search and logic by bifurcating statutory analysis into distinct, sequential tasks, which includes two relevant "Tasks":

Statutory Information Retrieval. Systems must scan a vast civil code corpus to retrieve all candidate provisions relevant to a specific legal query.

Statutory Textual Entailment. Systems must perform rigorous logical deduction to determine whether the retrieved provisions logically entail or contradict the given query [4, 40].

Year after year, architectures deployed by winning teams consistently reject single-stage monoliths in favor of pipelined approaches. The consensus among top-performing systems is that Information Retrieval requires high-recall Dense Retrievers (e.g., BM25 paired with Bi-Encoders) to handle corpus volume, while the entailment task strongly demands deep-reasoning Cross-Encoders. For instance, Althammer et al. [4] demonstrated during COLIEE 2021 that delegating the 'reading' phase to a disentangled Cross-Encoder completely prevents the semantic collapse inherent in long-document vectorization. This sustained global validation underscores that separating Information Retrieval (IR) from Natural Language Inference (NLI) is not just a hardware workaround, but the fundamental mathematical blueprint required to solve Semantic Conflict Detection as demonstrated in the Competition.

2.7.4 Pipeline Synthesis

Addressing context limits, logical blindspots, and morphological noise simultaneously is computationally impossible for a single monolithic model operating on consumer-grade LGU hardware. Therefore, this study adopts a dual-stage Coarse-to-Fine pipeline. Rather than a speculative architectural gamble, Reimers and Gurevych [104] note that this decoupled Retrieve-Then-Entail paradigm represents the empirical consensus in contemporary computational law.

The mathematical realities of Transformer scaling dictate this functional separation. Foundational work by Karpukhin et al. [63] established that Dense Retrievers must be independently optimized for Maximum Inner Product Search (MIPS) to handle corpus-level volume effectively. However, because Bi-Encoders lack the cross-attention necessary for deductive reasoning, they must feed into a secondary mechanism. Louis and Spanakis [79] validated this architectural split specifically for statutory law. They proved that utilizing a high-capacity vector retriever purely for filtering, followed by a localized Cross-Encoder for Semantic Conflict Detection, dramatically reduces false positives compared to single-stage retrieval systems.

Crucially, the necessity of this architectural decoupling is implemented by over a decade of evaluations from the Competition on Legal Information Extraction and Entailment (COLIEE). As the premier international benchmark for legal NLP, COLIEE structurally enforces the separation of search and logic by bifurcating statutory analysis into distinct, sequential operations: Statutory Information Retrieval and Statutory Textual Entailment [40]. Operating within this rigid competitive framework, Althammer et al. [4] demonstrated that in legal entailment tasks, delegating the "reading" phase to a disentangled Cross-Encoder prevents the semantic collapse inherent in long-document vectorization. Prior studies by Shao et al. [116] consistently corroborate this, showing that isolating the computational heavy-lifting of document filtering from the granular scrutiny of logical evaluation successfully bypasses the hardware bottlenecks that cause single-model architectures to fail.

Synthesizing these specific architectural components provides a rigorously tested blueprint: Stage One (Information Retrieval) absorbs the corpus volume and context length, while Stage Two (Textual Entailment) resolves the logical and morphological noise. This deliberate division of labor ensures semantic consistency across the unstructured archives of Philippine local legislation.

2.8 Information Retrieval

Information Retrieval (IR) is the process of extracting relevant documents from massive, unstructured datasets based on a specific query. In legal informatics, IR acts as the foundational filtering mechanism. It serves as the "coarse" stage of a dual-pipeline system, isolating candidate statutes from thousands of irrelevant municipal records before any intensive logical analysis takes place.

2.8.1 Nature of 'Relevance' in Retrieval

To understand how computational systems extract documents, we must first examine the human cognitive process that defines relevance in the physical world. In cognitive science, relevance is not a static property but a dynamic relationship between information and a specific task. According to Relevance Theory, humans naturally filter information by seeking the greatest cognitive benefit for the least mental effort [120]. When a legal researcher checks for conflicts, they do not simply count matching words; they instinctively evaluate the "intent" and "applicability" of a rule based on their background knowledge of the legal system. This human-centric view suggests that a document is relevant only if it provides a new piece of information that changes the researcher's understanding of the legal landscape [114].

Beyond the cognitive baseline, the definition of relevance in this study is anchored in the philosophical requirements of a functional legal system. According to the jurisprudence established by Lon Fuller in *The Morality of Law*, a primary requirement for the internal morality of law is consistency; a system that issues contradictory rules fails to be a legal system at all [36]. Within this philosophical framework, relevance is not merely about finding documents that talk about the same subject matter, but about identifying provisions that share a "normative overlap." This means that a retrieved document is relevant if it occupies the same jurisdictional or behavioral space as the proposed ordinance, potentially creating a situation where a citizen cannot obey one law without violating another. By grounding the retrieval stage in this quest for consistency, the system acts as a guardian of the legal system's internal logic, treating relevance as the identification of potential threats to the rule of law [36].

However, traditional computational methods struggle to replicate this human judgment because they rely on a rigid, mathematical definition of relevance. In standard deterministic systems, relevance is treated as a binary mathematical property based entirely on set intersection [14]. If we denote a candidate document as D and a user query as Q, this lexical relevance is formally evaluated as a binary function $R(D, Q) \in \{0, 1\}$. Under this strict lexical definition, a document is deemed fully relevant (i.e. 1) if and only if it contains the exact sequence of characters requested by the user [131]. Consequently, this mechanism establishes a purely topical baseline, effectively grouping documents together simply because they share a surface-level vocabulary [14]. While this mathematical simplicity allows for rapid indexing across massive databases, it completely strips away the contextual nuance required for complex legislative analysis.

This stark discrepancy between superficial lexical matching and actual legal consequence represents a major stumbling block for legacy search architectures. In statutory law, two documents discussing the exact same topic might be lexically relevant to one another, but they are only *legally* relevant if one provision actively modifies, permits, restricts, or preempts the other [131]. For instance, a retrieved document might use identical vocabulary but apply to a completely different jurisdiction or regulatory scope, rendering it completely useless for an ex-ante conflict check. Conversely, a document that governs the exact same civic behavior but uses entirely different phrasing—such as substituting "motor vehicle" for "automobile"—will be scored as mathematically irrelevant (i.e. 0) by a strict Boolean system [14]. Therefore, to accurately audit proposed Davao City ordinances, the retrieval system must abandon this binary, keyword-driven definition of relevance in favor of a dense, semantic approach capable of measuring actual jurisprudential relationships [67].

To resolve these limitations, modern computational pipelines fundamentally redefine relevance spatially rather than lexically. Instead of scanning a query as a raw string of text, the proposed system utilizes a Bi-Encoder architecture to compress entire legal provisions into high-dimensional mathematical coordinates, commonly referred to as dense vectors [104]. Under this advanced semantic framework, relevance is no longer determined by counting shared characters, but by calculating the geometric distance—such as cosine similarity—between these coordinate points [67]. If a newly drafted local ordinance and an existing national statute govern the same underlying civic behavior, the neural network learns to map their respective vectors closely together in the mathematical space, completely bypassing the vocabulary mismatch problem [144]. Consequently, the program evaluates relevance as a continuous proximity score rather than a rigid binary output, allowing it to successfully retrieve conceptually intertwined laws that a basic keyword search would otherwise ignore.

2.8.2 Long-Context Information Retrieval

Relying on standard BERT-based architectures introduces an artificial constraint on legal analysis: the 512-token limit. Because standard Self-Attention mechanisms scale quadratically with sequence length, processing long texts requires severe truncation. To get around this, researchers like Chalkidis et al. [22] routinely use Sliding Window techniques to segment documents into overlapping chunks. However, applying this to Philippine ordinances—which are historically wordy and often bury exceptions deep within 1,000-token provisions—is structurally destructive. It severs the crucial semantic link between a primary rule and its distant exception.

Consequently, the model operates with a logical blindspot. Beltagy et al. [11] note that this causes logical hallucinations, where a rule might be flagged as an absolute prohibition simply because its exempting clause was pushed into a separate, unconnected processing window. Furthermore, these systems must navigate the trade-off between Sparse Retrieval (e.g., BM25) for exact keyword matching, and Dense Retrieval for capturing conceptual meaning even without direct word overlap.

Resolving this fragmentation means abandoning chunking protocols entirely. The architectural solution requires an initial retrieval mechanism capable of native long-context retention. Recent Transformer models leverage advanced positional encoding, such as Rotary Positional Embeddings (RoPE), alongside the sparse attention matrices. Warner et al. [140] demonstrated that these upgrades extend context windows up to 8,192 tokens without triggering memory failures. Deploying this class of architecture allows the system to read entire local ordinances as cohesive semantic units, preserving the law's internal structure and eliminating the Sliding Window fallacy.

2.8.3 IR Benchmarks for Candidate Selection

Selecting the optimal retrieval architecture from a crowded field of open-weight models requires objective filtering. Rather than relying on architectural novelty, this study evaluates established empirical leaderboards to select candidates. While these testing frameworks mostly evaluate highly curated, Global North datasets, their rigorous zero-shot and clustering parameters provide a mathematically sound baseline. This helps identify models theoretically capable of handling the unstructured noise of Philippine municipal archives.

Benchmarking Information Retrieval (BEIR) The Benchmarking Information Retrieval (BEIR) framework evaluates a Dense Retriever's capacity for zero-shot retrieval across diverse and specialized datasets [124]. Zero-shot retrieval measures a model's ability to operate in a specific domain without prior fine-tuning. Because Philippine local legislation completely lacks annotated training sets, a candidate model must inherently possess robust zero-shot generalization. Thakur et al. [124] establish that a dominant position on the BEIR leaderboard is a critical proxy for this. It indicates the model's underlying capacity to map novel, localized legal vocabulary, including archaic syntax and uncodified administrative terms, into a coherent vector space without previous exposure.

Massive Text Embedding Benchmark (MTEB) Expanding on BEIR, Muennighoff et al. [87] introduced the Massive Text Embedding Benchmark (MTEB), which serves as the global standard for ranking embedding systems across over 130 datasets and multiple languages. For candidate selection, this study prioritizes top-performers within MTEB's "Retrieval" and "Clustering" subsets. Models excelling on these specific leaderboards have a proven capacity to group overlapping concepts and retrieve relevant texts based purely on semantic proximity. Therefore, MTEB acts as a foundational filter, ensuring that only architectures with empirically validated retrieval logic enter the noisy environment of Philippine statutory auditing.

2.8.4 Candidate Model Architectures for IR

The following retrieval models were selected based on their architectural advantages and performance on the MTEB leaderboard:

BAAI/bge-m3 Chen et al. [23] highlight BGE-M3 for its versatility in simultaneously performing Dense, Sparse, and Multi-Vector (ColBERT-style) retrieval. It provides a unified model foundation capable of processing varying inputs, from short sentences to long documents up to 8,192 tokens, across more than 100 languages [23]. Crucially, it obtains token weights similar to the BM25 algorithm without additional computational costs during dense embedding generation, making it a robust, production-ready baseline for large-scale municipal codes.

Qwen3-Embedding-8B Built on the Qwen3 foundation, the Qwen Team [101] positioned this model as the state-of-the-art upper bound on the MTEB multilingual leaderboard with a score of 70.58. Unlike traditional BERT-style encoders, it utilizes an auto-regressive language model architecture specifically adapted for embedding tasks. This allows it to excel at long-text understanding and instruction-following retrieval [101]. By successfully following complex relevance definitions, it delivers superior retrieval and clustering performance compared to earlier open foundation models.

Nomic-Embed-v1.5 Nussbaum et al. [95] introduced Nomic-Embed-v1.5 as the first fully auditable, long-context embedder to exceed OpenAI's text-embedding-ada-002 on the MTEB benchmark. It implements Matryoshka Representation Learning (MRL), a technique that produces resizable vectors ranging from 64 to 768 dimensions. As Nussbaum et al. [95] note, this allows developers to trade off embedding size for speed and memory savings with negligible performance loss. This flexibility is critical for scaling high-volume legal databases under tight computational constraints.

GTE-Multilingual Zhang et al. [149] designed this distilled encoder model to provide a joint semantic space for multiple languages while significantly reducing the computational burden of deployment. Distilled models typically retain over 95 percent of their teacher's performance, making them highly practical for real-world systems requiring high throughput and low latency [149]. Its specific ability to align different languages into a common vector space is essential for detecting semantic contradictions between the English and Cebuano versions of Davao City ordinances.

GTE-ModernBERT Warner et al. [141] present ModernBERT as a modernization of the traditional encoder-only Transformer. It integrates architectural upgrades like Rotary Positional Embeddings (RoPE) and Flash Attention to maintain high performance across an 8,192-token context window [141]. The model leverages an alternating attention mechanism that switches between local windowed attention for efficiency and global attention for long-range dependencies. According to Warner et al. [141], this achieves inference speeds over three times faster than standard BERT variants, making it ideal for time-sensitive legal applications.

2.9 Natural Language Processing

Natural Language Processing (NLP) provides the computational techniques needed to analyze and represent text at various linguistic levels, aiming for human-like comprehension [76]. As Jurafsky and Martin [61] explain, it acts as the bridge between human language and machine logic, allowing algorithms to parse and interpret unstructured text. In the legal field, NLP shifts document auditing from slow manual reviews to scalable, automated systems. However, applying standard NLP to law requires more than basic grammar checks; it demands systems that can navigate dense sentence structures, deontic operators (obligations, permissions, prohibitions), and the highly specific vocabulary found in statutory law.

Early Legal NLP relied heavily on Symbolic Artificial Intelligence (AI) [65]. These systems used manually built ontologies and strict Boolean logic to extract legal information. However, because municipal drafting language changes constantly over decades, these exact-match frameworks proved far too brittle. Contemporary NLP architectures solve this by moving to continuous vector representations. By mapping textual provisions into high-dimensional semantic spaces, modern Transformers bypass superficial vocabulary mismatches to mathematically quantify the deeper concepts connecting different legal terms. This allows the system to act as an active diagnostic engine that structurally maps legal conflicts, rather than just a passive search index.

This section breaks down how NLP operates within legislative analysis. It covers the systemic challenges of processing localized legal text, the structural need for Natural Language Inference (NLI), the benchmarks used for model selection, and the specific architectures built for these deductive tasks.

2.9.1 Challenges in NLP

Deploying NLP frameworks against municipal archives quickly exposes the mechanical limits of standard Language Models. These architectures must simultaneously overcome the geographic bias built into their pre-training data and the mathematical flaws of relying purely on lexical overlap.

The Global North Bias There is a noticeable lack of research addressing the noisy, low-resource reality of local government data in the Global South. As Ranathunga et al. [103] point out, the vast majority of Legal NLP benchmarks, like LexGLUE [22] and CaseHOLD, are trained exclusively on pristine, digitally native Western legal text. These highly optimized models degrade drastically when applied to Philippine Local Government Unit (LGU) archives. These datasets are filled with scanned PDFs suffering from Optical Character Recognition (OCR) errors, archaic American colonial phrasing, untranslated Spanish legal terms, and local "Taglish." Rust et al. [110] demonstrated that standard WordPiece tokenizers shatter these unrecognized words into meaningless sub-tokens, destroying the model's ability to infer legal intent. Fixing this localized morphological noise requires abandoning low-capacity tokenizers. Instead, developers must deploy models equipped with massive Byte-Pair Encoding (BPE) vocabularies (e.g., 128,000+ tokens) that can absorb linguistic quirks without triggering Out-of-Vocabulary (OOV) errors. Furthermore, He et al. [51] note that utilizing models with Disentangled Attention separates the actual content of a corrupted word from its noisy positional data. This allows the architecture to track logic accurately even when the text is heavily degraded by OCR artifacts.

Shortcomings of NLP in Lexical Similarity While early NLP systems excel at finding topical overlap, they remain structurally deficient for Semantic Conflict Detection. Traditional NLP heavily prioritizes Lexical Similarity—measuring the surface-level token intersection between documents using algorithms like TF-IDF or shallow word embeddings. Bowman et al. [16] explain that this paradigm is fundamentally unsuited for detecting legal contradictions because lexical metrics are entirely blind to deontic logic and syntactic negation. For example, an ordinance stating, "The city permits the sale of fireworks," and a later one asserting, "The city strictly prohibits the sale of fireworks," share near-perfect Lexical Similarity scores. A standard NLP similarity model will cluster them as highly related, completely missing the absolute policy collision happening beneath the vocabulary.

2.9.2 From NLP to Natural Language Inference

Historically, checking legislative consistency relied on Symbolic NLP (hand-written rules) and, later, Statistical NLP. In Statistical NLP, models learned probabilistic weights for specific inputs, like Term Frequency-Inverse Document Frequency (TF-IDF) or N-grams [65]. These models depended heavily on human-engineered features to define grammar rules and categories. While they allowed for soft, probabilistic decisions, they consistently struggled with the deep lexical and semantic ambiguities inherent in legal drafting.

The field largely transitioned to Neural NLP models in the mid-2010s [100]. Unlike statistical systems that treat features as separate entities, neural networks operate end-to-end, mapping raw text directly to output predictions through hidden layers. These architectures capture deeper linguistic nuances than traditional statistical methods by learning complex relationships during optimization. However, while machine learning excels at finding patterns, it can overly mimic its training data, making domain-specific pre-training essential for high performance in rigorous legal tasks.

Within this neural paradigm, Natural Language Inference (NLI) has emerged as the definitive benchmark for evaluating a model's capacity for actual reasoning. Yang et al. [146] define NLI as determining the logical relationship between two text sequences: a "premise" and a "hypothesis." The model must classify whether the hypothesis is true (Entailment), false (Contradiction), or unrelated (Neutral) based solely on the premise. This specific task exposes the core flaw of earlier statistical NLP, which mistakenly equated high word overlap with logical entailment [146]. Modern neural approaches fix this by capturing deep compositional semantics. As Le Hong et al. [72] point out, this allows modern models to resolve complex linguistic phenomena—like coreferences and modality—that rigid rule-based systems historically failed to grasp.

2.9.3 NLI Benchmarks

Identifying the optimal Cross-Encoder for the inference stage requires isolating models with robust, pre-trained deductive logic. Rather than relying on arbitrary accuracy metrics, this study uses established Natural Language Inference (NLI) leaderboards to filter candidates. While these benchmarks primarily evaluate Western jurisprudence, Chalkidis et al. [22] note that a model's baseline performance on these standardized tests is the most reliable indicator of its capacity to parse dense statutory syntax.

GLUE and SuperGLUE Wang et al. introduced the General Language Understanding Evaluation (GLUE) [138] and its more rigorous successor, SuperGLUE [137], as foundational baselines for model reasoning. These frameworks evaluate a Transformer's ability to handle diverse linguistic challenges, such as Recognizing Textual Entailment (RTE) and resolving complex coreferences. While a high SuperGLUE score confirms baseline cognitive capacity, it does not independently guarantee success in the legal domain due to the severe vocabulary shifts present in statutory drafting.

LexGLUE To evaluate specialized legal logic, candidate models are measured against the Legal General Language Understanding Evaluation (LexGLUE) benchmark. Chalkidis et al. [22] designed LexGLUE specifically to test a model's ability to navigate "legalese," where common words assume highly context-dependent meanings (e.g., "executed" denoting a signed contract rather than a penal death). A model's performance on LexGLUE's entailment tasks acts as a critical proxy for its ability to mathematically weigh legal conditions and exceptions. This proves the architecture has the baseline deductive framework necessary to handle the noisy, uncodified environment of Philippine municipal archives.

2.9.4 Candidate Model Architectures for NLI

The following inference models were selected based on their capacity for deep semantic reasoning and robustness on established logic benchmarks:

XLM-RoBERTa Conneau et al. [27] introduced XLM-RoBERTa as a foundational model for multilingual representation, trained on 2.5TB of text across 100 languages. Nie et al. [91] recognize it for its impressive zero-shot cross-lingual transfer capability, significantly outperforming earlier models on benchmarks like XNLI and XTREME. This architecture is vital for Davao City, allowing the system to project logical relationships across English and Cebuano to ensure local translations remain consistent with the source text.

RoBERTa-ANLI Adversarial NLI (ANLI) is a rigorous dataset designed to intentionally "break" models by exploiting their reliance on simple lexical heuristics [91]. RoBERTa models trained on ANLI demonstrate significantly higher robustness against "trick" inputs, such as premise-hypothesis pairs with high word overlap but contradictory meanings. This ensures the legal conflict detection system performs genuine logical inference instead of just recognizing surface patterns.

FLAN-T5 FLAN-T5 is an instruction-tuned encoder-decoder architecture leveraging over 1,800 diverse task templates to unlock reasoning capabilities for arbitrary instructions. Niklaus et al. [92] detail its specialized legal adaptation, FLawN-T5, which achieved a 50 percent improvement on the LegalBench suite by training on the massive LawInstruct dataset. This framework is highly effective because it can generate an explicit "explanation trace" for its logical predictions, helping human reviewers understand exactly why an ordinance was flagged as a conflict.

DistilBART DistilBART is a compressed variant of the BART architecture, using a "shrink and fine-tune" paradigm to reduce parameters while retaining language understanding. Shleifer and Rush [118] note it is 60 percent faster than full-scale counterparts and excels at zero-shot text classification, treating labels as natural language hypotheses for entailment checking. Its high throughput makes it an excellent candidate for real-time checking during active legislative drafting sessions, where low latency is critical.

DeBERTa-v3 DeBERTa-v3 achieved a landmark milestone as the first single model to surpass human performance on the SuperGLUE benchmark (scoring 89.9 vs. human 89.8). He et al. [51] explain that its Disentangled Attention mechanism processes token content and relative position information using separate vectors. This allows it to capture the subtle syntactic nuances necessary to distinguish legal "rights" from "obligations." By resolving the gradient "tug-of-war" between pre-training objectives, DeBERTa-v3 offers superior sample efficiency and precision for complex Semantic Entailment tasks.

2.10 Ground Truth

In supervised machine learning, a Ground Truth (or Gold Standard) dataset provides the correct, human-verified labels used to train and test models. During training, the neural network uses this dataset as an answer key. It constantly compares its own predictions against these true labels, calculating the difference (or loss) and adjusting its internal weights to make fewer mistakes over time [42]. Without a highly accurate set of target labels, the model will essentially learn the wrong rules, leading to a phenomenon where it memorizes flawed data instead of learning actual reasoning patterns.

Beyond just teaching the model, the ground truth is the only objective way to measure success. Researchers typically split this data into training, validation, and testing sets. The test set acts as the final exam, kept completely hidden from the model until the very end. Evaluating the model against this unseen ground truth allows researchers to calculate standard performance metrics like precision, recall, and F1-score. Without this baseline, it is mathematically impossible to prove that a model generalizes well to new data or works as intended in a real-world scenario [15].

2.10.1 Ground Truth in Natural Language Inference

In Natural Language Inference (NLI), ground truth datasets define the logical relationship between two pieces of text. Annotators read a source text (the Premise) and a claim (the Hypothesis), then categorize their relationship into one of three distinct labels. If the premise guarantees the hypothesis is true, it is labeled as Entailment. If they contradict each other, it is labeled as Contradiction. If there is not enough information to prove either, the pair is labeled Neutral [146].

Most traditional NLI datasets, such as the widely used SNLI (Stanford Natural Language Inference) corpus, are built using crowdsourced workers who write simple variations of everyday sentences [15]. While this approach is cheap and easily generates hundreds of thousands of training pairs, the resulting datasets are incredibly basic and often embed unintended annotation artifacts [48]. They usually feature short, single-sentence premises with obvious logical connections. Training a model exclusively on these simple datasets leaves it entirely unprepared for the dense, multi-layered logic required in specialized fields.

2.10.2 Annotation Procedures and Reliability Metrics

Turning raw text into a useful NLI dataset requires strict guidelines to minimize human bias and prevent annotators from relying on subjective legal interpretations. In standard NLI, guidelines dictate simple factual agreement. However, legal conflict detection requires evaluating normative relationships based on deontic logic. Drawing from the frameworks established by Agotnes et al. [2] and Li et al. [75], the annotation guidelines strictly instruct experts to map the text into obligations (must do), permissions (may do), and prohibitions (cannot do). A "Contradiction" is only flagged if the local ordinance mandates an action the national statute prohibits, or prohibits an action the national statute explicitly permits. If a local ordinance merely adds a regulatory step (e.g., a local processing fee) to a national permit without banning the core activity, the guidelines strictly dictate a "Neutral" classification. This rigid deontic framework guarantees that the dataset measures actual legal preemption rather than just topical friction.

To evaluate the effectiveness of these strict guidelines, researchers must utilize Inter-Rater Reliability (IRR) metrics [85] rather than mathematically flawed percentage agreements. Raw percentage agreement fails to account for statistical probability, where in a multiple-choice classification task, there is an inherent likelihood that annotators will occasionally select the same label by pure random guessing.

The two primary statistical formulas used to subtract this random chance are Cohen's Kappa [26] and Fleiss' Kappa [34]. Cohen's Kappa evaluates agreement between exactly two raters using the following formula:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where p_o represents the relative observed agreement between the two individual raters, and p_e represents the hypothetical probability of chance agreement.

When a dataset requires a multi-expert consensus—such as utilizing three or more domain experts to establish a majority rule—Fleiss' Kappa becomes the mandatory statistical metric. Fleiss generalized Cohen's formula to accommodate a fixed, larger number of multiple raters:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$$

In this generalized equation, $\bar{P}$ represents the mean observed agreement across all pairs of raters, while $\bar{P}_e$ represents the expected chance agreement across the entire pool. The resulting κ score is interpreted using the universally accepted statistical matrix established by Landis and Koch [71]:

Table 1: Interpretation of Fleiss' Kappa (κ) Scores

Kappa (κ) Score	**Interpretation**
< 0.00	Poor agreement
0.01 – 0.20	Slight agreement
0.21 – 0.40	Fair agreement
0.41 – 0.60	Moderate agreement
0.61 – 0.80	Substantial agreement
0.81 – 1.00	Almost perfect agreement

While objective tasks routinely achieve κ scores above 0.90, applying this expectation to the legal domain is mathematically unrealistic. Bowman et al. [15], in their foundational study for the Stanford Natural Language Inference (SNLI) corpus, established that evaluating semantic relationships is inherently subjective. Their benchmark study hit an approximate performance ceiling with an overall Fleiss' Kappa of 0.70. They explicitly demonstrated that achieving near-perfect agreement (> 0.80) in textual inference usually indicates overly simplistic or fabricated test data.

2.10.2.1 Mitigating Sub-Threshold Reliability

While achieving a "Substantial" Inter-Rater Reliability score ($\kappa > 0.61$) validates the clarity of the annotation guidelines, failing to meet this threshold does not inherently invalidate the dataset or halt the experimental pipeline. Klebanov and Beigman [68] establish that a sub-threshold κ score typically indicates systematic ambiguity within the text itself—categorized as "hard" instances—rather than incompetence by the annotators. Their research demonstrates that it is still entirely possible to extract a reliable, low-noise subset from the instances where annotators do achieve consensus.

When researchers encounter this sub-threshold reliability, the mitigation strategy depends entirely on the size of the dataset. For large-scale studies, including the disaster-tweet classification by Ablazo [1], the standard procedure is data omission. Beigman et al. [68] explicitly advocate for filtering out difficult, borderline cases from the gold standard, arguing that evaluating algorithms on inherently ambiguous text introduces statistical noise akin to random coin-flips. In these massive datasets, items that fail to reach majority consensus are simply discarded because the volume of remaining "easy" data easily absorbs the loss.

However, applying this standard omission strategy to a specialized, few-shot legal dataset mathematically destroys the test set's Minimum Detectable Effect. Furthermore, in municipal legislation, these "hard" cases often represent the exact jurisdictional friction the logic engine needs to learn. To preserve dataset volume without introducing benchmarking noise, researchers in specialized domains employ Expert Adjudication rather than omission. Gao et al. [39] establish that in these scenarios, a hierarchical resolution protocol is activated where a senior domain expert reviews the conflicting annotations. The senior expert resolves the ambiguity based on overarching jurisprudential principles and manually establishes the Gold Standard label, treating the initial low κ metric as a measure of text difficulty rather than a rigid experimental blocker.

2.10.3 Ground Truth in the Legal Domain

When applying NLI to the legal domain, the standard approach of comparing short, isolated sentences quickly breaks down. Statutory phrasing is notoriously complex, filled with conditional clauses, exceptions, and domain-specific vocabulary. Consequently, a hypothesis in a legal context rarely aligns perfectly with a single sentence in the premise. Instead, proving an entailment or contradiction often requires the model to connect evidence scattered across multiple paragraphs or pages.

To address this, researchers have developed specialized legal datasets that reflect real-world document structures. For instance, the Competition on Legal Information Extraction/Entailment (COLIEE) dataset constructs its ground truth by pairing civil code statutes with legal bar exam queries, requiring models to identify specific legal provisions before determining entailment [102]. Similarly, the ContractNLI dataset by Koreeda & Manning [69] engineers its gold standard around non-disclosure agreements. Rather than giving the model pre-cut sentence pairs, ContractNLI forces the system to evaluate hypotheses against entire legal documents, requiring annotators to both classify the logical relationship and extract the specific supporting text span.

Furthermore, unlike general-domain datasets crowdsourced through platforms like Amazon Mechanical Turk, legal ground truth relies heavily on domain experts. Constructing datasets like CaseHOLD [150], which evaluates legal holding statements, requires law students and legal professionals to ensure the labels accurately reflect actual jurisprudence. The push towards comprehensive legal benchmarks, such as LexGLUE [22], demonstrates that evaluating legal NLP models demands rigorously structured, expert-annotated data rather than automated or amateur labeling.

This difficulty is magnified when dealing with local or municipal legislation. While national or federal laws generally follow standardized drafting conventions, municipal texts are highly heterogeneous, exhibiting significant variations in terminology, structure, and overall organization across different jurisdictions [17]. Because of this localized variance, municipal legal documents have historically been neglected in NLP research, hampered primarily by a severe lack of annotated ground truth datasets [18]. Standard legal corpora fail to capture the specific legislative templates, localized governance terminology, or idiosyncratic cross-referencing styles used in municipal laws, such as Philippine local ordinances. Without existing resources that reflect these structural and linguistic nuances, evaluating models on local legislation requires building highly specialized, expert-annotated datasets from the ground up.

A critical architectural decision in dataset construction is determining the exact granularity of the extracted legal premise. While the hypothesis (the local draft) is evaluated as a multi-sentence span to capture context, the premise (the national statute) must be strictly isolated at the sentence level. In Philippine statutory drafting, a single paragraph frequently contains a general rule, multiple conditional modifiers, and distinct exceptions. Holzenberger et al. [53] demonstrate that feeding an entire statutory paragraph into an NLI model as a premise dilutes the model's attention mechanism, causing it to lose track of specific variable constraints. By forcing human annotators to isolate the national rule down to its exact sentence boundary, the resulting Ground Truth prevents the candidate logic engines from getting distracted by surrounding, non-governing text, ensuring the classification mathematically targets the exact point of semantic collision.

The difficulty of dataset construction is severely magnified by the "expert bottleneck" inherent to legal informatics. As Zhong et al. [151] explicitly note, the complex terminology required for legal texts makes manual labeling highly time-consuming and difficult to scale, as it strictly requires domain experts rather than crowdsourced workers. Because constructing massive datasets for localized municipal law is practically unfeasible, researchers must rely on constrained, highly curated Gold Standards. To compensate for this limited data volume, the current NLP standard mandates treating legal conflict detection as a few-shot learning scenario. Gao et al. [38] validate this approach, proving that transformer models can be successfully fine-tuned on specialized datasets containing merely hundreds of examples, provided the evaluation framework rigorously controls for overfitting.

2.10.4 Multi-Sentence Context and Cross-Document Limits

Traditional Natural Language Inference (NLI) evaluates text using a strict 1-to-1 paradigm, comparing a single premise sentence directly against a single hypothesis sentence. However, this highly granular approach fundamentally breaks down in the legal domain due to two specific edge cases. First, intra-document context: a primary rule in one section of an ordinance is frequently modified or negated by an exception buried several paragraphs later (the "1v2" or multi-sentence problem). Second, cross-document dependencies: a contradiction may only emerge when synthesizing provisions from two entirely separate statutes simultaneously.

To address the first edge case, modern legal frameworks utilize Asymmetric Span-Level Pairing. Yin et al. [147] critique traditional sentence-level datasets, proving that capturing contextual exceptions requires evaluating multi-sentence blocks rather than isolated lines. As demonstrated by Koreeda and Manning in the ContractNLI dataset [69], the premise remains a single, focused legal rule, while the corresponding hypothesis is extracted as a multi-sentence chunk. This asymmetric structure allows the model's attention mechanism to accurately evaluate a primary rule against a distant exception contained within the same local provision.

While asymmetric spans successfully capture intra-document context, resolving the second edge case—cross-document multi-hop reasoning—remains computationally prohibitive. Identifying semantic preemption that spans multiple independent laws currently represents the bleeding edge of NLP research and is not yet fully explored for local governance. Recent frameworks have attempted to solve this, but they introduce severe architectural overhead. For instance, Lu et al. [80] address cross-document retrieval by programmatically mapping Entity Graphs to mine evidence paths between laws, a process that is highly resource-intensive and unscalable for unstructured LGU archives. Alternatively, Shen et al. [117] proposed the HopWeaver framework, which relies on massive generative Large Language Models (LLMs) to synthesize complex reasoning paths across documents.

Both of these state-of-the-art solutions immediately violate the strict hardware constraints of consumer-grade LGU workstations. Consequently, practical, edge-deployable systems must intentionally bound their theoretical scope to single-premise, asymmetric span evaluations, deliberately avoiding the quadratic memory scaling required for multi-document fusion.

2.10.5 Train/Test Partitions for Small Datasets

When evaluating NLI models on constrained datasets, researchers often default to k-fold cross-validation to maximize testing coverage. However, Vabalas et al. [128] mathematically demonstrated that applying cross-validation to datasets under 1,000 items introduces severe optimistic bias, causing models to perform significantly worse in real-world deployment than their test scores suggest. To rigorously control for overfitting in few-shot legal scenarios, contemporary evaluation frameworks must abandon k-fold approaches. Instead, they require strict, static Train/Validation/Test partitions (e.g., 70/15/15) to ensure the evaluation metrics remain entirely untainted by training exposure.

Beyond merely preventing optimistic bias, utilizing a static partition allows for rigorous validation monitoring during the fine-tuning phase. In datasets with limited volume, neural networks are highly susceptible to memorizing the training data rather than learning underlying logical patterns. Goodfellow et al. [42] establish that permanently isolating a validation set allows architectures to employ early stopping protocols, halting the training process the moment the model's accuracy on unseen data begins to degrade. Furthermore, as Zhang et al. [148] note regarding few-sample fine-tuning, this static separation ensures strict document-level boundaries, guaranteeing that clauses from a specific municipal draft do not accidentally bleed across the training and testing sets to artificially inflate performance metrics.

2.10.6 Small Test Sets and Minimum Detectable Effect

A standard critique of static splits on small datasets is that the resulting test set lacks the statistical power to reliably compare candidate architectures. However, Card et al. [19] established that a test set's validity is fundamentally determined by its Minimum Detectable Effect (MDE) rather than arbitrary size thresholds. The MDE dictates that if the expected accuracy gap between two models is small, a massive test set is required to prove the difference is not just luck. Conversely, because fine-tuning foundational models on a highly novel, localized municipal domain yields massive margins of performance improvement, the required MDE drops significantly. Consequently, even a constrained test set of approximately 50 pairs is mathematically sufficient to definitively detect the optimal logic engine without sacrificing statistical significance [38].

The practical application of MDE fundamentally shifts how model performance is evaluated in specialized domains. In general NLP research, massive test sets are required because researchers are often evaluating models against benchmarks like GLUE to prove marginal, incremental gains of one percent to two percent over previous state-of-the-art architectures [138]. However, applying a base foundation model—which possesses no prior knowledge of Philippine local governance for example—to a highly specialized task generally results in baseline performance barely above random chance [20]. Zheng et al. [150] demonstrate that in the legal domain, fine-tuning foundational architectures on expertly curated pairs triggers a massive, double-digit jump in classification accuracy compared to zero-shot baselines. Because this expected performance delta is so large, the system does not need thousands of examples to prove the improvement is statistically valid, such that the 50-item test set is perfectly calibrated to mathematically capture this distinct leap in the model's reasoning capacity.

2.11 Explainable Artificial Intelligence

In legal informatics, a ”black box” prediction is unacceptable, regardless of its accuracy. A system that flags a conflict between two ordinances must provide the reasoning (*ratio decidendi*) behind its classification to genuinely aid legislative analysis. Without clear justification, human reviewers cannot trust or legally defend the algorithm’s output. Consequently, the proposed architecture integrates Explainable AI (XAI) not as an afterthought, but as an intrinsic component of the inference process. This approach prioritizes visualizing the model’s actual decision path over generating simplified text summaries.

2.11.1 Intrinsic Explainability Mechanisms

The architecture relies on Intrinsic Explainability via Attention Mechanisms rather than Surrogate Models like LIME or SHAP. These external explainers are computationally expensive and inherently prone to approximation errors because they attempt to learn a separate, simplified linear model around the main prediction. Instead, Attention Weights, or the numerical values the model assigns to specific tokens during inference, can be directly extracted to generate ”soft” explanation heatmaps. This native extraction ensures that the explanation perfectly matches the mathematical operations the network actually performed. By bypassing secondary approximations, the system guarantees a much higher degree of structural transparency.

While using attention as a direct proxy for explanation has sparked vigorous debate, it remains the most practical metric for text-based analysis. Jain and Wallace [60] famously argued that Attention Weights do not always correlate perfectly with feature importance. However, Wiegreffe and Pinter [142] provide a crucial counter-argument, contending that Attention Mechanisms offer a ”faithful” representation of the model’s internal prioritization when rigorously tested. In the context of Semantic Conflict Detection, this specific property allows the system to highlight the exact phrases in Ordinance A (e.g., ”absolute ban”) and Ordinance B (e.g., ”conditional permit”) that triggered the logical contradiction. This creates a verifiable visual link between the input text and the output classification, delivering transparency without the massive latency cost of running a separate explanation model.

Complementing this visual transparency is the critical implementation of Confidence Calibration. Deep Learning models, particularly modern Transformers, frequently exhibit extreme overconfidence in incorrect predictions when processing out-of-domain text. As Guo et al. [47] demonstrate, neural networks are inherently poorly calibrated, meaning their unadjusted Softmax probabilities are mathematically deceptive and fail to reflect true predictive uncertainty. To build a reliable risk assessment metric, the proposed system actively mitigates this flaw by applying post-hoc calibration, specifically Temperature Scaling, to the network’s pre-Softmax logits. Systematically rescaling these logits before normalizing them successfully aligns the output confidence scores with the actual likelihood of correctness. This crucial transformation replaces illusory confidence with mathematically sound metrics, allowing users to safely filter results between a ”clear conflict" (e.g., a calibrated 99 percent probability) and a "potential ambiguity" (e.g., a 55 percent probability).

2.11.2 Edge Deployment Mechanisms

The operational reality of a Local Government Unit (LGU) demands strict adherence to hard hardware and latency limits. Post-hoc explainability methods operate by repeatedly perturbing input data, which requires hundreds or thousands of forward passes per document just to approximate feature importance [106, 82]. Recent empirical evaluations by Kumar et al. [70] confirm that permutation-based explainers increase inference latency by orders of magnitude, making them computationally prohibitive for lengthy legal texts. Furthermore, Wang et al. [139] highlight that deploying heavy post-hoc XAI pipelines on local Edge Servers routinely degrades throughput entirely below acceptable thresholds for real-time analysis. Consequently, adopting Intrinsic Explainability methods, such as extracting native Attention Weights and Softmax logits, is the only viable path to achieve transparency without crashing consumer-grade local hardware.

2.11.3 Metrics for Explanation Quality

Evaluating explainability requires carefully balancing two often-competing metrics: Plausibility and Faithfulness. Plausibility measures how convincing and human-readable the explanation appears to the end user. Conversely, Faithfulness measures how accurately the explanation reflects the model's actual internal reasoning process. Jacovi and Goldberg [59] warn that optimizing purely for Plausibility often leads to misleading explanations that actively mask model biases or hallucinations. In this study, priority is strictly and exclusively given to Faithfulness to ensure legal reliability.

Therefore, the evaluation framework entirely rejects "smoothed" or "generative" explanations that might rewrite the law just to sound simpler. These auto-generated summaries introduce a dangerous layer of interpretation that can subtly alter legislative intent. Instead, the system presents raw Attention Scores directly derived from the mathematical inference layers. While these heatmaps may be noisier to read, they truthfully reveal exactly which tokens the model utilized to reach a verdict of Entailment or Contradiction. This transparent approach aligns perfectly with the rigorous standards of legal scholarship, where the accuracy of the citation always takes precedence over the elegance of the prose.

This strict adherence to Faithfulness directly informs the system's rejection of Generative Explainable AI. Recent trends in legal tech utilize Large Language Models (LLMs) to automatically generate plain-English summaries explaining why a conflict occurred. However, Linna [77] warns that generative models are notoriously prone to "hallucinations," frequently fabricating plausible but entirely fake legal citations or misrepresenting statutory intent when summarizing complex provisions. Jacovi and Goldberg [59] classify these text summaries as highly "plausible" but dangerously "unfaithful," as the generated text does not reflect the actual mathematical operations the model used to classify the text. By restricting the XAI output to raw, intrinsic Attention Heatmaps, the proposed architecture functions as a strict "white-box" system. It does not attempt to explain the law; it visually proves exactly which tokens triggered the contradiction flag, entirely eliminating the risk of generative hallucinations in the final legal audit.

2.11.4 XAI within the Dual-Stage Pipeline

Integrating Explainable AI into a Retrieve-Then-Entail architecture requires deliberately isolating the explanation generation to the second stage of the pipeline. The initial Information Retrieval phase functions purely as a high-throughput, opaque filter designed to handle massive corpus volume. As Anand et al. [5] detail, attempting to extract feature importance during this initial retrieval phase is computationally wasteful and provides little analytical value to the end user. Conversely, the Cross-Encoder stage performs the rigorous Natural Language Inference (NLI) needed to formally confirm a semantic conflict. This makes the secondary stage the most needed and most logical site for generating explainability metrics.

Within this second stage, the system would leverage Raw Computational Transparency instead of outsourcing the explanation to an external text generator. Rather than relying on a Generative LLM to summarize the contradiction, which often prioritizes human plausibility over factual faithfulness [59], the architecture extracts the raw Attention Weights and pre-Softmax logits directly from the Cross-Encoder's internal layers. For the end-user interface, this raw computational data must be mapped cleanly back to the source text to be useful. By overlaying the highest-weighted tokens as visual heatmaps directly onto the text of the ordinances, the system provides immediate, visual proof of the conflict's exact origin [134]. This interface strategy ensures that legislative reviewers do not have to blindly trust a black-box prediction; they can literally see the mathematical decision path the model took to classify the relationship as a contradiction.

2.12 Theoretical Framework

As explored in the earlier sections, the Retrieve-then-Entail architecture has emerged as an ideal theoretical paradigm within the domain of legal informatics, particularly for handling complex statutory reasoning and conflict detection. As demonstrated in the proceedings of the Twelfth International Competition on Legal Information Extraction and Entailment (COLIEE 2025) [41], state-of-the-art systems strategically bifurcate the problem into two distinct computational phases. The initial phase demands a high-recall information retrieval mechanism to rapidly sift through massive legal corpora, while the subsequent phase requires high-precision natural language inference to determine exact logical relationships between the retrieved texts. This theoretical framework establishes a two-phase architecture. It combines a highly scalable, dense Information Retrieval (IR) pipeline to isolate relevant statutory context, followed by a lightweight, encoder-based Natural Language Inference (NLI) pipeline for rapid conflict detection. This structure adapts the methodologies of Team AIIR Lab for statute law retrieval and Team KIS for legal rationale extraction.

2.12.1 Statute Law Retrieval by Team AIIR

The theoretical foundation for the first phase of this system is grounded in a Dense Information Retrieval (IR) architecture, heavily adapted from the top-performing, encoder-based methodologies utilized in modern legal text processing, specifically drawing from the AIIR Lab's approach to statute law retrieval [144]. The primary theoretical objective of Phase 1 is to bridge the semantic gap between a newly submitted document and a massive corpus of existing laws (e.g., 10,000+ documents) without relying on computationally expensive, real-time text generation or manual graph construction. To achieve this, the framework is divided into four core theoretical components: Data Augmentation, Semantic Vectorization, Corpus Pre-computation, and Live Retrieval.

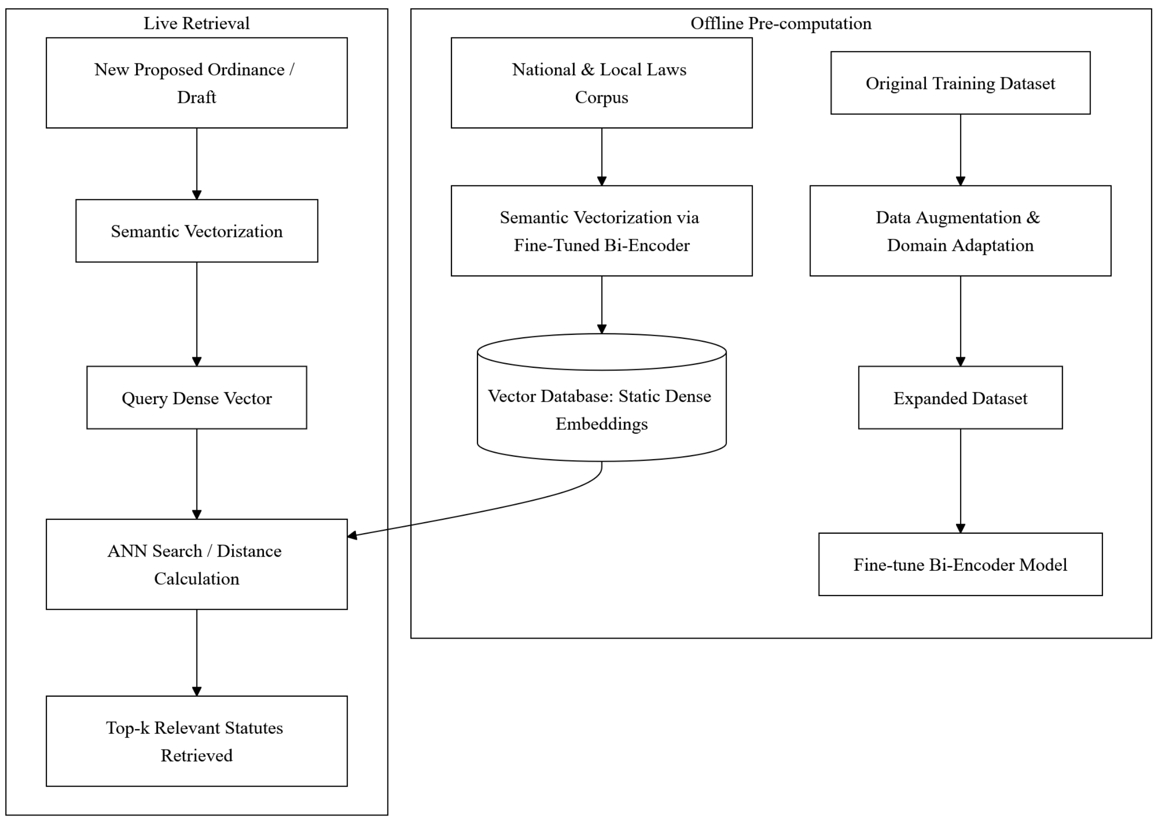

Figure 1: Phase 1 - Statute Law Retrieval (Bi-Encoder Pipeline). Adapted from AIIR Lab's retrieval methodology.

Data Augmentation and Domain Adaptation. By training on this artificially expanded dataset, the retrieval model learns to recognize the same core legal concepts regardless of whether they are expressed in complex statutory jargon or simplified modern text, drastically improving the system's semantic robustness.

Semantic Vectorization via Bi-Encoder. The core engine of Phase 1 is a Bi-Encoder Architecture (such as the all-mpnet-base-v2 model used by AIIR Lab), which maps text into a dense and high-dimensional mathematical space. Unlike cross-encoders, which must read the query and the document simultaneously and are too slow to run across thousands of documents in real-time, a bi-encoder processes the query and the candidate documents independently. The augmented training dataset from the previous step is used to fine-tune this bi-encoder. During fine-tuning, the model learns to adjust its embedding space so that vectors representing a query and its truly relevant legal articles are pulled closer together, while irrelevant pairs are pushed further apart. Optimal model performance is theoretically achieved by evaluating the embeddings using distance metrics such as cosine similarity, Euclidean distance, and Manhattan distance.

Corpus Pre-computation and Storage. In order to satisfy the constraint of executing searches across thousands of documents efficiently, the framework employs an offline pre-computation strategy. Before the system is ever deployed live, the fine-tuned bi-encoder processes the entire corpus of existing national and local laws. It converts every single law, article, and provision into a static, dense vector embedding. These embeddings are stored in a highly scalable vector database. Because the vectorization of the corpus is decoupled from the live search process, the system can effortlessly scale to tens of thousands of documents without requiring heavy real-time GPU compute.

Live Retrieval and Distance Calculation. The final component represents the live user interaction. When a new text (e.g., a proposed ordinance) is submitted to the system, the bi-encoder instantly converts this single document into a dense vector using the same mathematical dimensions as the pre-computed corpus.The system then performs an approximate nearest neighbor (ANN) search, calculating the mathematical distance (e.g., cosine similarity) between the newly submitted vector and the 10,000+ stored vectors. Because this relies entirely on mathematical distance calculation rather than generative text decoding, the system retrieves the top-k most relevant statutes in milliseconds. This purely dense retrieval methodology has been proven to achieve high recall and efficiency, seamlessly narrowing down a massive corpus to a highly focused subset of relevant candidates, perfectly priming the data for the second phase of fine-grained logical analysis.

2.12.2 Statute Law Retrieval by Team JNLP

This framework outlines a highly optimized, three-stage IR pipeline designed to extract relevant statutory articles from a massive legal corpus. Modeled after Team JNLP's methodology for statute law retrieval which achieved the highest F2 score [41], this architecture focuses on progressively narrowing down the search space through three main phases: Pre-Retrieval, Model Fine-Tuning, and Result Ensemble.

2.12.2.1 Pre-Retrieval

The primary objective of the first phase is to drastically reduce the expansive search space while maintaining a near-perfect recall rate. This is achieved through a two-step filtering process that drops low-relevance articles.

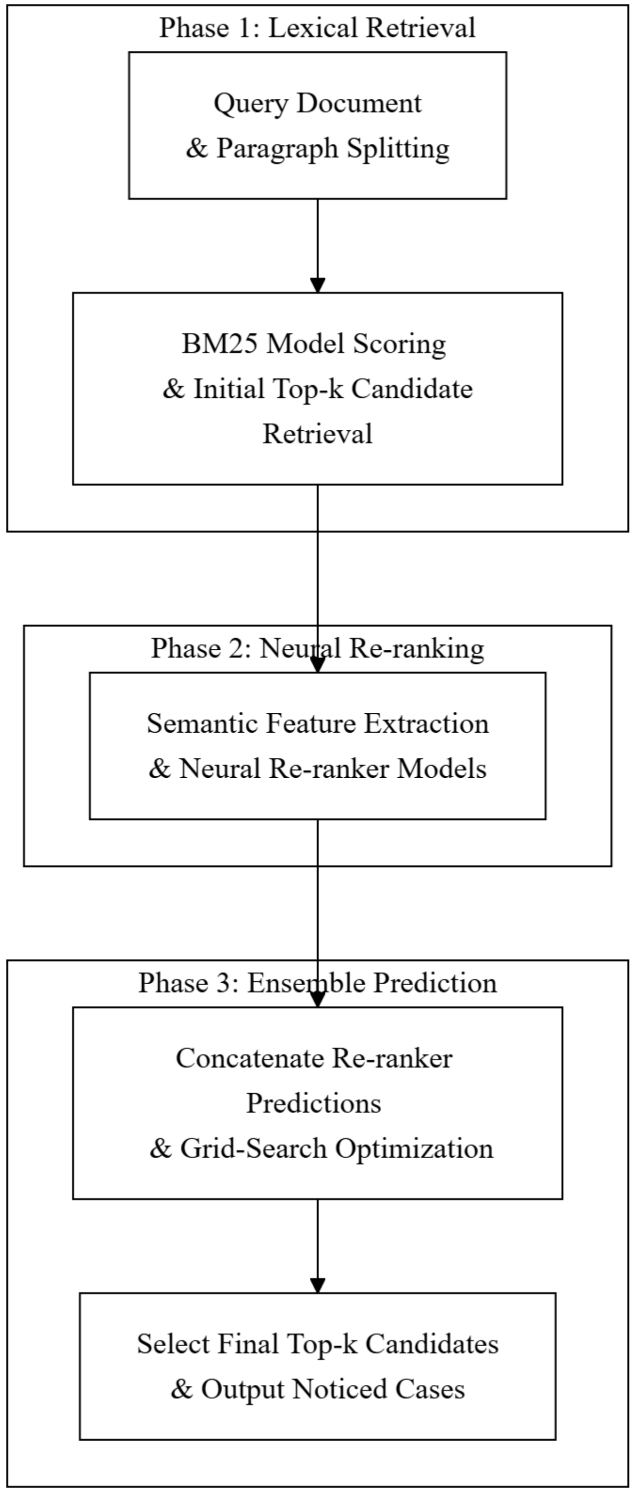

Figure 2: Phase 1 - Statute Law Retrieval (Bi-Encoder Pipeline). Adapted from JNLP's retrieval methodology.

Distance-Based Heuristic Filtering. Dense text vectors for both the proposed query and the corpus articles are extracted using a robust embedding model (e.g., BGE-M3). Instead of relying on direct cosine similarity, the framework calculates the L1 distance between the query vectors and all article vectors, organizing these distances into a predetermined number of bins (e.g., 76 bins). A Gradient Boosting classifier is then trained on these binned distance features to predict relevance. To counter severe class imbalance, the positive (relevant) query-article pairs are heavily over-sampled (e.g., replicated 300 times). The system retains only the top 100 candidate articles with the highest predicted probability.

Passage Re-ranking and Pruning. The initial pool of 100 candidates is further refined to remove the least relevant documents. A specialized passage re-ranker model (i.e. RankLLaMA) evaluates the candidates and prunes the bottom 50, successfully narrowing the search space down to the top 50 most relevant articles without causing a catastrophic drop in overall recall.

2.12.2.2 Model Fine-Tuning

Once the search space is constrained to the top 50 candidates per query, the framework applies a suite of state-of-the-art Large Language Models (LLMs) to perform fine-grained binary classification (determining if a query-article pair is relevant or not).

Parameter-Efficient LLM Adaptation. Because fine-tuning billion-parameter models is computationally prohibitive, the framework employs Parameter-Efficient Fine-Tuning (PEFT) techniques, specifically LoRA and QLoRA. These techniques allow massive causal language models and text embedding models to learn the deep semantic relationships of the legal text with minimal computational overhead.

Class Imbalance Mitigation. The dataset fed into this stage is inherently imbalanced, as only one or two articles out of the 50 candidates are actually relevant. To ensure the models learn to properly identify positive matches, an up-sampling strategy is applied again during training, replicating the positive samples (e.g., 3 times) to balance the ratio of relevant to irrelevant pairs.

2.12.2.3 Result Ensemble

To maximize the robustness of the retrieval system and mitigate the biases of any single LLM, the final phase aggregates the predictions from the batch of models fine-tuned in the previous phase.

Weighted Sum Aggregation. Rather than relying on simple majority voting, the system calculates the final relevance of a legal article by computing a weighted sum of the probability scores outputted by the various fine-tuned LLMs.

Hyperparameter Optimization. The weight assigned to each individual model's prediction is not static, as it is dynamically tuned and optimized using an automated hyperparameter optimization framework, namely Optuna. This mathematical ensembling ensures the highest possible retrieval accuracy by perfectly balancing the strengths of the diverse models in the pipeline.

2.12.3 Statute Law Textual Entailment by Team KIS

The theoretical foundation for the second phase of the system functions as a Natural Language Inference (NLI) pipeline, shifting from broad document retrieval to precise logical evaluation. To maintain high execution speeds and minimize computational overhead, this phase utilizes a lightweight, non-generative encoder architecture, specifically modeled after the ModernBERT-based rationale extraction methodology by KIS [62].

This phase is composed of four theoretical components:

Unified Input Construction. Traditional encoder models face bottlenecks due to short sequence limits, which force long legal documents to be heavily segmented. This framework overcomes that limitation by leveraging the extended context window of modern encoders, which can support up to 8,192 tokens. The retrieved existing law (the premise) and the newly proposed provisions (the hypotheses) are concatenated into a single, continuous text string. Special tokens are utilized to cleanly separate the different legal segments within this unified sequence, ensuring the model understands the boundaries of the premise and the hypothesis.

Token-Level Formulation. To evaluate specific potential conflicts within the proposed text, the framework applies a granular formatting strategy. Special separation tokens (e.g., ¡sep¿) are strategically inserted immediately preceding each individual provision or claim being evaluated within the unified string. This transforms the raw legal text into a highly structured mathematical input tailored for targeted classification.

Simultaneous Binary Classification. The unified sequence is processed by a highly efficient, lightweight encoder (e.g., a 130-million parameter model). Because it is an encoder-only architecture, it completely bypasses the computational overhead of decoding and generating new text. Instead, it performs direct mathematical classification. The model treats each strategically placed $< sep >$ token as an independent classification head. It simultaneously outputs a binary probability score for each provision, indicating whether the proposed text logically conflicts with or is entailed by the existing retrieved law.

Probability Aggregation and Ensembling. In order to prevent overfitting and ensure the highest degree of reliability in its legal reasoning, the framework concludes with an ensemble decision mechanism. Multiple versions of the encoder are trained using diverse hyperparameter configurations (such as varying epochs and learning rates) or different subsets of the training data. During live execution, the predicted probability scores from the top-performing models in the ensemble are aggregated. If the averaged probability across the ensemble exceeds a predefined threshold (e.g., > 0.5), the system outputs a definitive classification flagging a semantic conflict.

2.12.4 Theoretical Significance

This part summarizes how the theories discussed in this chapter will be applied to the development of the conflict detection system. While the previous sections explained these

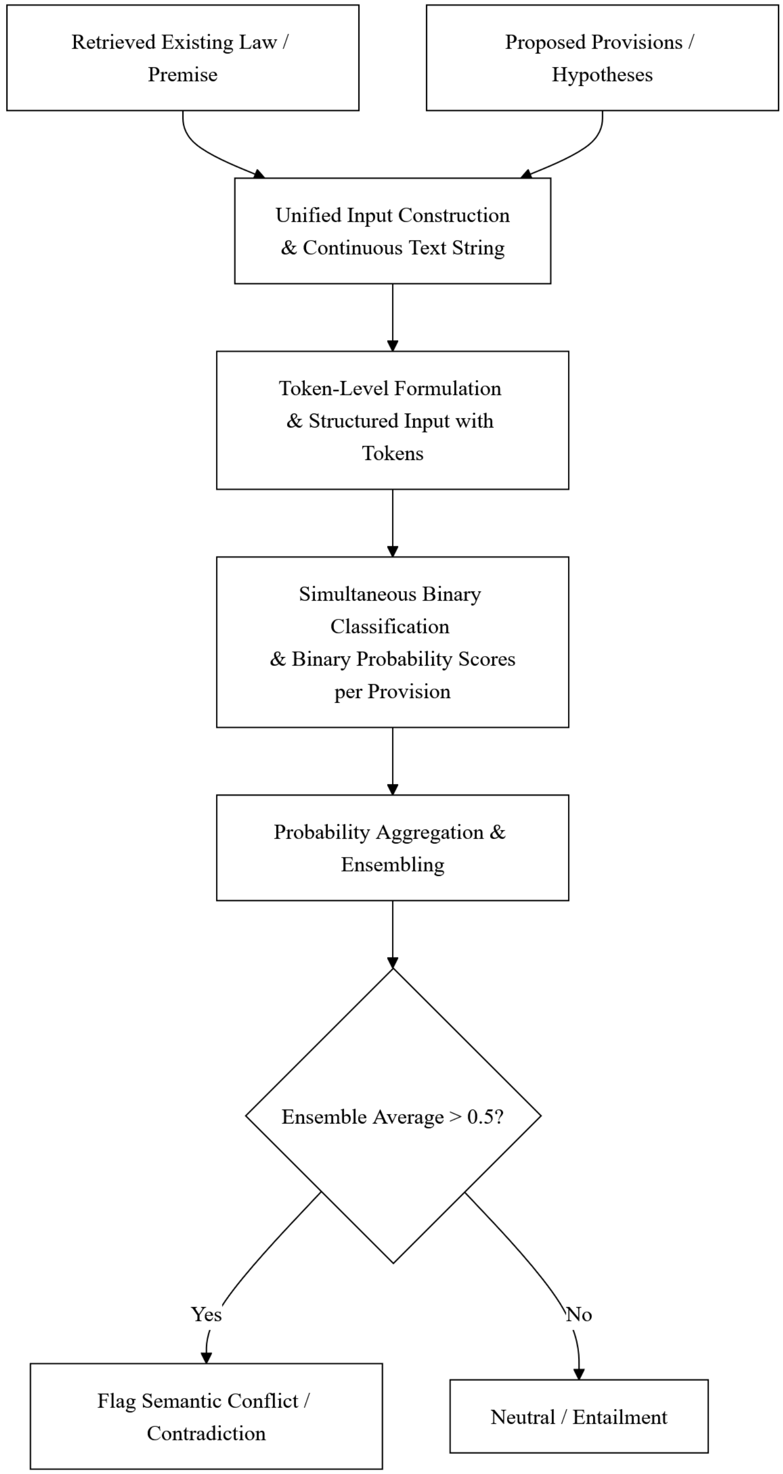

Figure 3: Phase 2 - Statutory Textual Entailment (NLI Pipeline). Adapted from Team KIS's ModernBERT rationale extraction methodology.

models as general research, this synthesis itemizes the specific parts of each framework that will serve as the engine for the Davao City system.

Dense Retrieval Methodology. The study adopts the dense retrieval methodology from the **AIIR Lab** to handle the initial processing of laws [144]. Specifically, the system uses their approach to ”Data Augmentation” to fix OCR errors in scanned Davao City ordinances. By following their strategy of ”Semantic Vectorization,” the system will turn raw legal text into mathematical coordinates (embeddings) that can be searched quickly on the proponents’ local hardware [144].

Ensemble Learning Strategy. To ensure the system is both fast and accurate, the study applies the ensemble strategy used by **Team JNLP** [40]. This study adopts their two-step ”Lexical and Neural” search process for Stage 1. First, a simple keyword search (BM25) will remove thousands of unrelated laws. Second, their method of ”Weighted Sum Aggregation” will be used to combine search scores, ensuring the system picks the most relevant candidates without needing expensive cloud servers [40].

Textual Entailment. For the actual logic check in Stage 2, the study implements the **Team KIS framework** for textual entailment [62]. This study borrows their ”Unified Input Construction” method, which combines the draft and the existing law into one long string (up to 8,192 tokens) for the AI to read. Following their ”Binary Classification” strategy, the model will judge multiple rules at the same time to see if they logically clash, which is the most efficient way to detect conflicts on standard computers [62].

Synthesis. By combining these three frameworks, the proponents are not just building a search tool, but an automated ”diagnostic triage” engine. The AIIR Lab framework provides the clean data, the JNLP framework provides the search speed, and the KIS framework provides the deep logic reasoning. This integrated approach allows the Davao City Sangguniang Panlungsod to perform ex-ante audits that were previously too slow or too difficult to do manually.

Chapter 3

Methodology

This chapter details the technical methodology used to build and evaluate the Coarse-to-Fine Semantic Conflict Detection System. It covers how the Philippine legal text is gathered and cleaned, how the two-stage Retrieve-then-Entail architecture is structured, and how the models are selected. It also outlines the evaluation metrics, the creation of the ground truth dataset, and the specific hardware constraints of the proponents, which are intended to mimic the standard computing capacity of an average Local Government Unit (LGU).

3.1 Conceptual Framework

This section maps how data flows through the proposed system. The architecture follows a basic Input-Process-Output (IPO) logic, broken into a two-step "Coarse-to-Fine" pipeline. By separating the task of searching for documents from the task of actually checking their logic, the system avoids the hardware bottlenecks that usually cause large language models to fail. The steps below outline how a raw document is processed into a final conflict alert.

3.1.1 Phase 1: Pre-Planning

This initial phase focuses on establishing the system's foundation by readying the offline database and finding the best logic model.

Corpus Processing. The system first compiles a knowledge base by extracting over 10,000 national laws and local scanned ordinances. Because local municipal archives suffer from heavy optical character recognition (OCR) noise, the data undergoes rigorous cleaning and standardization. The finalized texts are then converted into mathematical vectors (embeddings) and stored offline in a pre-computed vector database.

Model Training & Selection. To ensure the system evaluates logic accurately, candidate Cross-Encoder models compete against a curated, expert-annotated Ground Truth dataset. The model that demonstrates the highest accuracy and F1-Score

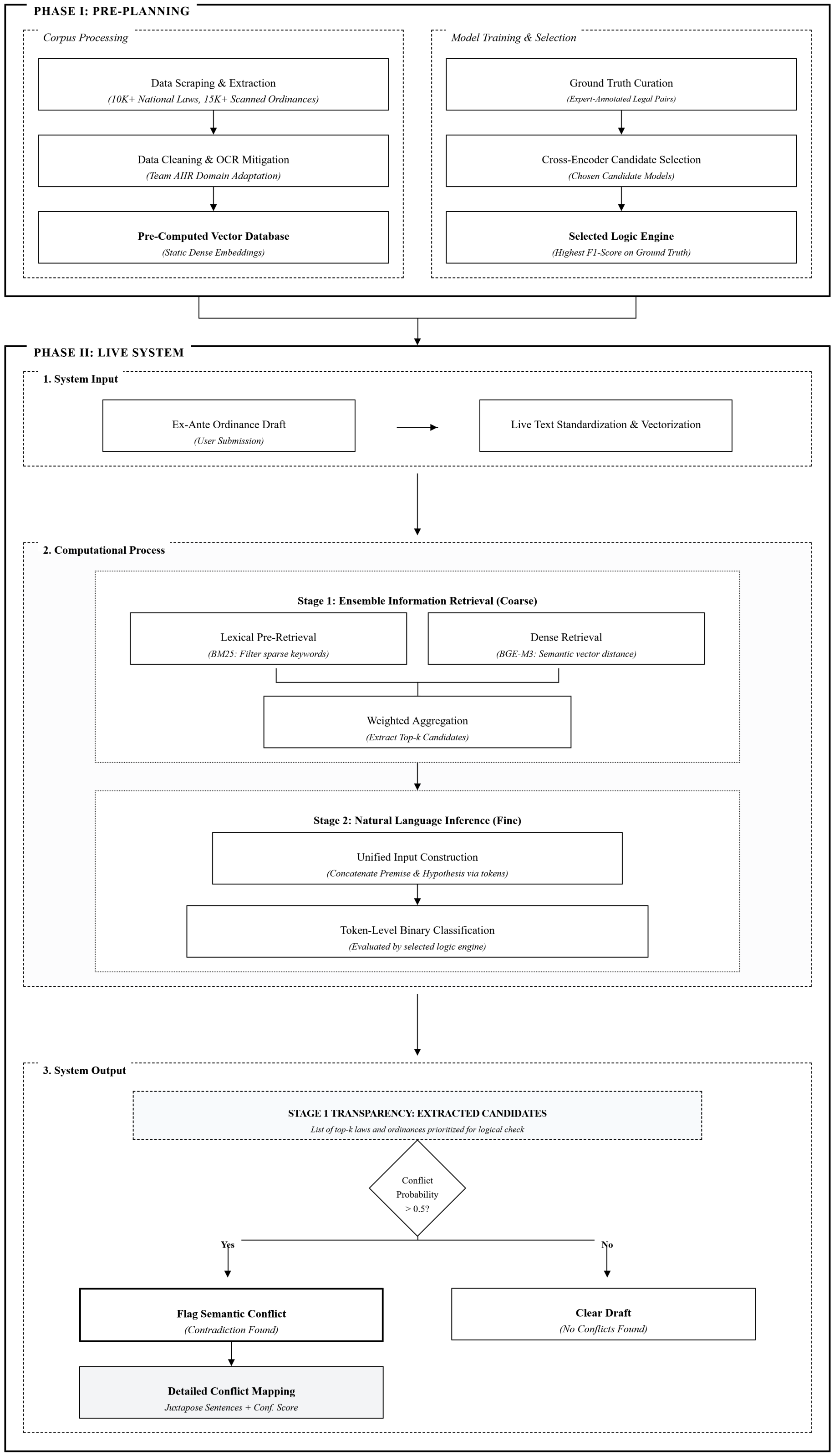

Figure 4: Conceptual Framework

in detecting contradictions is selected as the permanent logic engine for the live system.

3.1.2 Phase 2: Live System

The live system handles the actual consistency checking and is structured around an Input-Process-Output (IPO) pipeline.

System Input. The live workflow begins when the user submits a new, ex-ante ordinance draft into the system. The system immediately cleans, standardizes, and vectorizes this submitted text so it can be compared against the pre-computed database.

Computational Process. This part involves the two main stages: the information retrieval to retrieve relevant laws and the textual entailment to execute the conflict detection.

Stage 1: Information Retrieval. The system first acts as a broad search engine to rapidly reduce the massive database. It combines a basic lexical keyword search (BM25) from Team AIIR [144] to filter out unrelated laws with a dense retrieval model (Bi-Encoder) that measures semantic vector distance inspired from Team JNLP [41]. The scores are combined using weighted aggregation to extract only the top-k most relevant statutory candidates, which is the ensemble learning part derived from Team JNLP [41].

Stage 2: Natural Language Inference The pipeline then shifts to actual logical reasoning using the winning NLI model, implementing a framework specifically modeled after the Team KIS methodology [62]. The system creates a unified input sequence that concatenates the proposed draft provisions with the retrieved existing laws into one continuous string. Following the "Binary Classification" strategy derived from Team KIS [62], the lightweight encoder bypasses text generation and performs a simultaneous binary classification. It evaluates the unified sequence by treating specific separation tokens as independent classification heads, outputting a binary probability score for each provision to judge multiple rules at the same time and mathematically determine if a logical conflict exists.

System Output. In the final stage, the system delivers a definitive verdict based on the model's calculated conflict probability. If no conflicts are detected, the system officially clears the draft. If a contradiction is flagged, the system outputs a detailed conflict mapping that specifies exactly which part of the text is conflicting. To ensure transparency, it utilizes Explainable AI (XAI) to project an attention heatmap, explicitly highlighting the exact words in the documents that triggered the semantic conflict.

3.2 Methodology

This section outlines the comprehensive methodology governing the data collection, system engineering, and evaluation phases. It details the precise protocols used to build the pipeline from raw text acquisition to the final prototype demonstration.

3.2.1 Research Design

This research is fundamentally categorized as applied research designed to engineer a specialized software solution for civic governance. The experimental systems development approach is utilized to address the inherent complexities of the manual ordinance review bottleneck through automated detection. This explicit framing ensures methodological clarity regarding the integration of legal informatics and computational linguistic principles within a localized framework. By prioritizing the practical application of transformer-based architectures, the design bridges the gap between theoretical modeling and administrative deployment. The investigation serves as an empirical validation of the retrieve-then-entail pipeline's capability to operate within a specific regulatory environment.

The experimental design involves isolating critical computational modules to measure retrieval latency and classification accuracy independently. Iterative prototyping allows for the systematic optimization of the information retrieval and natural language inference stages against local statutory data. This structured engineering logic ensures that the final artifact is optimized for the specific semantic nuances of municipal and national legislative texts. Applied research methodologies facilitate the direct translation of high-level NLP concepts into a functional tool for the Davao City Sangguniang Panlungsod. Consequently, the research outcomes establish a scalable benchmark for automating ex-ante consistency checks in evolving legal corpora.

3.2.2 Data Acquisition and Compilation

The construction of the legal corpus requires the systematic extraction of text from disparate online repositories to ensure a high-recall baseline. These repositories host the documents identified as critical for evaluating the ex-ante ordinance consistency. To maintain linguistic consistency and simplify the vectorization process, the system strictly acquires the English-language versions of all legal texts. This scope aligns with Philippine statutory construction rules where the English text is prioritized in the event of a conflict with vernacular translations. National statutes and local ordinances are gathered using distinct programmatic pipelines tailored to the specific DOM structures of each source.

The compilation process involves the use of programmatic scraping tools to automate the retrieval of raw legal text. Headless browser automation and DOM parsing libraries are utilized to navigate the archival pagination of the selected websites. Once a document is identified, the script extracts the raw text and its associated metadata, such as the enactment date and the law's formal title. This raw data is then stored in a temporary database for deduplication and initial formatting. The following subsections detail the specific repositories utilized for national and local legislative data.

National Sources. The Lawphil Project is designated as the primary repository for national statutes due to its extensive collection and machine-readable text availability [7]. The website features a relatively static and straightforward HTML structure that facilitates reliable programmatic data scraping. Scripts are configured to navigate the alphabetical and chronological indexes to extract Republic Acts and Commonwealth Acts efficiently. Despite its breadth, the Lawphil archive occasionally exhibits data gaps regarding recently enacted legislation or specific obscure administrative orders. To address these deficiencies, the scraping protocol identifies missing numerical sequences for targeted retrieval from alternative online legal databases. Supplemental legal repositories are utilized as fallback mechanisms to acquire specific statutes not present in the primary dataset. These secondary sources provide the necessary redundancy required to maintain the integrity of the statutory corpus. Data retrieved from these additional archives undergoes the same JSON conversion and morphological cleaning as the primary Lawphil extraction. The system cross-references multiple versions of a law to ensure the most authoritative and up-to-date text is indexed in the vector database. Standardizing the format across all national sources allows for uniform semantic processing in the downstream retrieval pipeline.

Local Sources. The Legislative Information Support System Program (LISSP) serves as the primary data source for Davao City municipal ordinances. Following formal consultations with the Sangguniang Panlungsod (SP) staff, the proponents were granted access to the publicly available online LISSP database. This database contains the most comprehensive digital record of local legislation enacted by the city council of which are available digitally [24]. The acquisition strictly targets English-language ordinances to ensure a standardized vocabulary across the municipal and national datasets. This direct access to the municipal record ensures that the Information Retrieval phase operates on the same baseline used by human legal reviewers. However, in instances where specific ordinances are missing from the LISSP repository, the Davao City SP official website is utilized as a fallback. This secondary local source provides the necessary redundancy to ensure that the ex-ante check is performed against a complete legislative record. A significant technical challenge arises because the LISSP and SP archives primarily host scanned, image-based PDF files rather than native text documents. These files lack a searchable text layer, acting as the primary source of character-level noise within the dataset. The extraction of raw text from these visual artifacts represents a critical computational hurdle that must be resolved prior to semantic vectorization.

3.2.3 Data Preprocessing

The unstructured nature of the scraped local archives necessitates an aggressive data cleaning and normalization sequence. While national statutes are digitally native, the Davao City local ordinances act as the primary source of OCR-induced noise in the dataset. Because these local archives consist of scanned, image-based PDFs, the extraction process is prone to character misclassifications and erratic line breaks. The pipeline executes rule-based heuristics to identify and correct these artifacts specifically within the municipal ordinance subset. This targeted cleaning ensures that the subsequent semantic vectorization stage is not compromised by fragmented or unreadable text strings.

Following text normalization, the documents undergo structural chunking to comply with the sequence length constraints of the transformer models. This specific preprocessing step is modeled after the methodology established by Team AIIR for legal information extraction [144]. Large legal provisions are partitioned into semantically coherent segments using a sliding window algorithm with a defined overlap parameter. As outlined in the conceptual framework, this approach ensures critical legal context is preserved across provision boundaries. The resulting finalized text segments are then mapped into the vector database to support the dual-stage retrieval and inference pipeline.

3.2.4 Ground Truth Process

Evaluating the deductive logic of candidate Cross-Encoder models in Stage 2 necessitates a highly specialized ground truth dataset. Unlike the coarse retrieval phase, the Natural Language Inference (NLI) pipeline requires deterministic, expert-validated examples to fine-tune and benchmark its binary classification accuracy. Because existing benchmarks fundamentally lack Philippine municipal context, a custom answer key must be curated specifically for the Davao City jurisdiction. This curated dataset provides the absolute objective baseline for identifying semantic preemption between local drafts and national statutes. Consequently, this ground truth process bridges the critical gap between raw computational vectorization and applied legal logic.

3.2.4.1 Legal Categorization and Asymmetric Span Pairing

To construct a statistically representative ground truth, the raw statutory corpus must first undergo strict topical categorization. The researchers deploy an existing machine learning classification model, engineered during a prerequisite academic elective, to assign the documents into specific legislative domains like zoning or taxation. Following this categorization, the system isolates specific national statutory provisions at the granular sentence level to serve as the fixed premise. However, to capture multi-hop reasoning and cross-sentence dependencies, the system abandons strict 1-to-1 sentence pairing. Instead, it utilizes Asymmetric Span-Level Pairing, where the corresponding hypothesis extracted from the local ordinance is maintained as a multi-sentence block [147]. This asymmetric structure (1 premise vs N hypothesis sentences) ensures that contextual exceptions and referential pronouns within the local draft are evaluated holistically against the national rule [69].

The corresponding hypothesis spans paired with the premise sentences are compiled using a hybrid approach of historical extraction and adversarial text perturbation. Initially, the researchers extract actual historical amendments and vetoed drafts from the municipal archive to capture natural legislative clashes [58]. To ensure a balanced distribution of logical labels, annotators supplement the dataset by manually injecting negative constraints into existing spans, mirroring the perturbation strategies of Koreeda and Manning [69]. The inclusion of these human-engineered adversarial examples is statistically necessary to prevent the neural networks from relying on lazy lexical heuristics during inference [91]. These finalized, asymmetric premise-hypothesis pairs are then compiled into a structured evaluation instrument paired with strict coding guidelines for the human annotators.

While the system utilizes asymmetric spans to capture intra-document context, it explicitly excludes cross-document multi-hop reasoning (i.e., identifying contradictions that only emerge when synthesizing two or more distinct national laws). Resolving multi-premise dependencies currently represents the bleeding edge of natural language inference. State-of-the-art frameworks addressing this require either computationally expensive entity graph retrieval [80] or the use of massive generative models for synthetic path generation [117]. Implementing these N-vs-N architectures would immediately violate the strict consumer-grade hardware constraints established for Local Government Unit deployment. Consequently, the ground truth and subsequent inference pipeline are strictly bounded to single-premise evaluations.

3.2.4.2 Annotation Workflow and Reliability Metrics

Manual classification of the sampled text pairs requires significant domain expertise to ensure jurisprudential accuracy. The Davao City Sangguniang Panlungsod currently operates with 24 elected councilors comprising of eight councilors from each of the city's three districts [152]. After further clarification with the SP staff, the proponents discovered that the councilors are each supported by two dedicated legal researchers, creating a specialized talent pool of forty-eight professionals. Because interpreting local legislative preemption is their primary operational competency, the researchers will recruit fifteen of these active legal researchers to serve as the primary human annotators. These annotators are provided with the stratified sample pairs alongside explicit classification guidelines detailing the exact conditions required for an Entailment, Contradiction, or Neutral logical state.

Following the formal consensus protocol adapted from Ablazo [1], a logical classification is only accepted into the final Ground Truth if a minimum of two out of the three assigned experts agree. To mathematically validate this consensus across the groups, the system calculates Fleiss' Kappa to quantify the Inter-Rater Reliability (IRR). In instances where all three annotators provide conflicting labels—resulting in a complete absence of majority consensus—the item is not discarded. Instead, following the hierarchical adjudication protocols established for specialized domains by Gao et al. [39], a designated Senior Legal Researcher from the SP intervenes as the formal Adjudicator to definitively establish the Gold Standard label.

3.2.4.3 Batched Annotation and Consensus Protocol

To prevent annotation fatigue without compromising statistical reliability, this study directly adapts the batched distribution framework established by Ablazo [1]. In their baseline research for disaster-related text classification, Ablazo required an overlap of three independent evaluations per item to establish a valid majority consensus. To manage a dataset of 12,054 tweets, this required a total workload of over 36,000 individual annotations. By distributing this massive workload across a pool of 250 crowdsourced students, the required work per person was reduced to roughly 145 tweets. This established the precedent of using batched distribution to maintain a strict three-vote overlap without overwhelming the annotators.

This study applies that exact mathematical framework to the specialized legal domain. Because cross-referencing multi-sentence ordinances requires significantly more cognitive load than reading short social media text, the dataset size is smaller, but the distribution logic remains identical. The exact allocation for the domain experts is mathematically derived as follows:

$$\begin{aligned}
N &= 350 \text{ premise-hypothesis pairs} \\
k &= 3 \text{ independent votes per pair} \\
P &= 15 \text{ legal annotators}
\end{aligned}$$

$$\begin{aligned}
W &= N \times k \\
W &= 350 \times 3 \\
W &= 1,050 \text{ total annotations}
\end{aligned}$$

$$\begin{aligned}
\text{Work per person} &= \frac{W}{P} \\
\text{Work per person} &= \frac{1,050}{15} \\
\text{Work per person} &= \mathbf{70} \text{ pairs per annotator}
\end{aligned}$$

To execute this distribution practically, the dataset is partitioned into five equal blocks of 70 pairs each. The 15 researchers are divided into five distinct sub-panels of three experts, with each sub-panel independently evaluating a single block. This ensures every premise-hypothesis pair receives exactly three independent evaluations while keeping the individual workload highly manageable.

3.2.4.4 Inter-Rater Reliability Evaluation

To mathematically validate the quality of the Ground Truth, the system evaluates annotator consensus using Fleiss' Kappa (κ). While raw percentage agreement indicates how often the panel selected the same logical label, it fails to account for instances where annotators might agree by pure random chance. Fleiss' Kappa isolates the actual analytical consensus by mathematically subtracting the probability of random guessing from the final score.

This specific statistical metric is explicitly required over Cohen's Kappa. Cohen's formula is strictly limited to evaluating exactly two raters, whereas the proposed batched distribution assigns exactly three experts per premise-hypothesis pair. The resulting reliability score is interpreted using the standard evaluation framework established by Landis and Koch [71]. In their foundational statistical matrix, they explicitly delineate a κ score between 0.61 and 0.80 as the strict mathematical threshold for "Substantial Agreement."

Applying this specific baseline to natural language processing is empirically supported by Bowman et al. [16] in the creation of the Stanford Natural Language Inference (SNLI) corpus. Evaluating semantic relationships is inherently subjective; in their benchmark study, Bowman et al. established an approximate performance ceiling for NLI tasks with an overall Fleiss' Kappa of 0.70 (specifically scoring 0.77 for Contradiction, 0.72 for Entailment, and 0.60 for Neutral). They demonstrated that achieving near-perfect agreement (> 0.80) in textual inference is highly improbable and usually indicative of overly simplistic test data. Because Philippine municipal legislation contains denser syntax than general English sentences, expecting an Inter-Rater Reliability score higher than the SNLI 0.70 ceiling is statistically unrealistic. Therefore, the dataset targets a minimum κ threshold of 0.61. Surpassing this threshold officially classifies the expert consensus as "Substantial," mathematically proving that the rigid annotation guidelines successfully standardized the panel's legal reasoning.

3.2.5 Experimental Setup and Hardware Constraints

The physical execution environment is deliberately restricted to consumer-grade hardware to mirror the standard computing capacity of an average Local Government Unit. All model training, vectorization, and inference tasks are processed on a local machine equipped with a standard central processing unit and adequate graphical memory. This specific constraint simulates the realistic technological boundaries of the Davao City legislative offices. By anchoring the experimental setup to accessible hardware, the research guarantees the pipeline's deployability without requiring expensive cloud computing infrastructure. The system must operate efficiently within these boundaries to prove its practical viability.

Operating within these strict hardware limits mandates the explicit exclusion of massive Large Language Models (LLMs). Deploying generative architectures with billions of parameters would immediately exceed the available memory thresholds and incur unacceptable processing latency. Instead, the architecture strictly utilizes lightweight encoder-only models optimized for rapid sequence classification. This constraint forces the pipeline to rely on highly efficient vector math rather than computationally expensive autoregressive text generation. Consequently, the hardware limitations directly dictate the architectural shift towards the proposed Retrieve-then-Entail methodology.

3.2.6 Empirical Model Selection

Before the Natural Language Inference pipeline is deployed for live execution, the system must definitively identify the most accurate logic engine. The proponents execute an empirical selection phase to evaluate a pool of candidate Cross-Encoder architectures against the curated Ground Truth dataset. To satisfy the project's hardware constraints, the candidate pool is strictly limited to models that are relatively lightweight and memory-efficient. Despite their smaller size, these selected candidates must possess a proven baseline performance in complex natural language reasoning tasks. By isolating these models against a standardized, expert-validated baseline, the process objectively determines which architecture best understands local legislative preemption.

In order to ensure the empirical contest is statistically rigorous and reproducible, the 350 curated premise-hypothesis pairs are partitioned using a strict static split for the Train/Validation/Test of 70/15/15. This specific division yields 245 training pairs, 52 validation pairs, and 53 test pairs. Operating within this restricted sample threshold formally categorizes the evaluation as a few-shot fine-tuning scenario, a methodology proven highly viable for large language architectures [38]. While k-fold cross-validation is traditionally favored, Vabalas et al. [128] mathematically demonstrate that static train/test splits are actually mandatory for datasets under 1,000 items to prevent optimistic bias and strictly control overfitting. Furthermore, maintaining a 53-item test set satisfies the Minimum Detectable Effect (MDE) thresholds established by Card et al. [19], guaranteeing the test set possesses the statistical power needed to validate the architecture's reasoning improvements.

3.2.7 Model Implementation and Pipeline Execution

The live execution of the consistency checking system strictly follows a bifurcated architecture designed to balance rapid search speeds with deep logical accuracy.

Stage 1: Information Retrieval. The live pipeline initiates when the user submits an ex-ante ordinance draft. The system first triggers the BM25 lexical algorithm to rapidly scan the static corpus. This sparse retrieval phase acts as a computational filter, immediately discarding laws that share zero keyword vocabulary with the drafted text. Simultaneously, the surviving documents are processed by the Bi-Encoder, which maps both the draft and the corpus into a dense vector space. The Bi-Encoder calculates the mathematical cosine distance between these vectors to capture deep semantic relationships that simple keyword matching misses. Once both the sparse and dense scores are calculated, the pipeline executes a weighted ensemble aggregation. Following the methodology established by Team JNLP, the system mathematically normalizes the distinct scoring outputs to ensure they operate on a compatible scale. The normalized scores are then merged using a pre-defined weighting parameter to balance exact keyword matches with broader semantic relevance. This merged calculation generates a strictly ordered shortlist containing only the top-k most relevant statutory provisions. By restricting the output, the system effectively isolates the specific legal texts required for the subsequent deep logic review without overloading the hardware.

Stage 2: Natural Language Inference. Upon receiving the top-k shortlist from Stage 1, the pipeline transitions to rigorous logical evaluation using the Natural Language Inference (NLI) module. The system programmatically constructs a unified input sequence for each retrieved candidate provision. It concatenates the retrieved statutory text (functioning as the premise) and the drafted ordinance (functioning as the hypothesis) into a single string. To guide the attention mechanism of the selected logic engine, the pipeline injects strict structural markers, specifically placing a $[CLS]$ token at the sequence start and a $[SEP]$ token between the two legal texts. This formatting ensures the model mathematically recognizes the boundary between the existing law and the proposed rule. The formatted sequence is then fed into the pre-selected Cross-Encoder model for simultaneous processing. The model leverages disentangled attention mechanisms to evaluate the semantic dependencies between every token in the premise against every token in the hypothesis. It executes a binary classification over the entire sequence, calculating the precise logical relationship between the paired texts. The final layer of the architecture outputs calibrated softmax probabilities that explicitly label the relationship as an Entailment, Contradiction, or Neutral state. If the contradiction probability exceeds the designated safety threshold, the system triggers the Explainable AI (XAI) module to flag the provision for user review.

3.2.8 Evaluation Metrics

The predictive performance of the classification models is quantified using standard mathematical evaluation metrics. Precision is calculated as the ratio of true positive contradiction flags to the total number of flags generated by the system. Recall is computed as the ratio of true positive contradiction flags to the total actual contradictions defined in the Ground Truth. These specific measurements isolate the model's ability to minimize false alarms while successfully catching genuine legal conflicts. High precision ensures users are not overwhelmed with incorrect alerts, while high recall guarantees critical preemptions are not ignored.

To establish a single comparative baseline for the candidate models, the system computes the harmonic mean of precision and recall. This metric, defined mathematically as the F1-Score, provides a balanced assessment of the model's overall diagnostic accuracy. The calculation heavily penalizes extreme disparities between precision and recall, ensuring the selected model does not favor one metric at the expense of the other. The Cross-Encoder achieving the highest F1-Score on the annotated pairs is permanently integrated into the final pipeline. These rigorous mathematical evaluations validate the system's empirical reliability before deployment.

3.2.9 Prototype Development

The validated algorithmic pipeline is ultimately deployed as a lightweight, single-page web application to facilitate end-user interaction. The graphical user interface is deliberately engineered with minimal complexity, visually mirroring the straightforward architecture of standard online file conversion utilities. Users initiate the diagnostic process through a drag-and-drop upload zone that accepts the ex-ante ordinance drafts in standard document formats. Upon submission, the interface activates a real-time loading state to indicate that the dual-stage retrieval and inference pipeline is actively processing the text against the static corpus. This minimalist design eliminates extraneous navigational elements, ensuring the legislative staff can execute the semantic conflict check without requiring specialized technical training.

Upon completion of the pipeline execution, the system dynamically updates the interface to display a multi-tiered results dashboard. To ensure transparency in the Information Retrieval stage, the interface provides a dedicated section listing the top-k statutory provisions extracted during Stage 1. This "source transparency" module allows users to verify which national laws were prioritized as semantically relevant prior to the deep inference classification. If a conflict is flagged, the dashboard generates a granular error report explicitly juxtaposing the specific sentence in the submitted draft against the exact conflicting provision. The report includes a numerical confidence score derived from the logic engine, mathematically justifying why a particular section was identified as a contradiction.

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